Accelerating the Tower Number Field Sieve with Galois Automorphisms

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Abstract. The Number Field Sieve algorithm and its variants are the best known algorithms to solve the discrete logarithm problem in finite fields. When the extension degree is composite, the Tower variant TNFS is the most efficient one. Looking at finite fields with composite extension degrees such as 6 and 12 is motivated by pairing based cryptography that does not have yet a good quantum-resistant equivalent.
The two most costly steps in TNFS are the *relation collection* and the *linear algebra* steps. While the use of order k Galois automorphisms allows to accelerate the relation collection step by a factor of k, using them to accelerate the linear algebra step remains an open problem. This problem is solved for $k = 2$, leveraging a quadratic acceleration factor equal to 4.
In this work we bring a solution both for $k = 6$ and $k = 12$. We propose a new construction that allows the use of an order 6 (resp. 12) Galois automorphism in any finite field $\mathbb{F}_{p^6}$ (resp. $\mathbb{F}_{p^{12}}$), thus accelerating the linear algebra step with approximately a factor of 36 (resp. 144). Moreover, we provide a SageMath implementation of TNFS of our new construction, and validate our findings on small examples.

1 Introduction

Context. The discrete logarithm problem in a cyclic group $\mathbb{G}$ with a generator $g \in \mathbb{G}$ is the computational problem of finding an integer x modulo $|\mathbb{G}|$ for a given target $T \in \mathbb{G}$, such that $T = g^x$. Despite the growing interest in post-quantum cryptography, the discrete logarithm problem is still at the basis of many currently-deployed public key protocols. This article deals with the discrete logarithm problem in the group of invertible elements of a finite field, $\mathbb{G} = \mathbb{F}_{p^n}^*$, excluding small characteristic finite fields due to the existence of quasi-polynomial time algorithms [5, 16, 26].

The Number Field Sieve. The most efficient algorithm to compute discrete logarithms in finite fields is (a variant of) the Number Field Sieve (NFS) algorithm [15, 23]. All the variants of NFS have a subexponential complexity equal to $\exp((c + o(1)) \cdot (\log Q)^{1/3}(\log\log Q)^{2/3})$ where $o(1)$ tends to 0 as the finite field size Q tends to infinity and $c > 0$ is a constant that depends on the variant.

The Tower variant, TNFS[1] [24, 25, 34] applies when the extension degree n is composite and its asymptotic complexity is lower than NFS. De Micheli, Gaudry

[1] Sometimes referred to as the extended Tower Number Field Sieve (exTNFS).

and Pierrot [12] reported in 2021 the first implementation of TNFS and performed a record computation on a 521-bit finite field with extension degree $n = 6$. One year later Robinson [32] reported a record computation using TNFS on a 512-bit finite field of extension degree $n = 4$. These records confirm that TNFS is currently the best practical algorithm that computes discrete logarithms in composite extension degree finite fields. Table 1 compares the performance of NFS and TNFS. Note that the two TNFS-records are approximately 100 bits larger than those performed with NFS, and yet their cost are lower.

Year	Finite field	Bitsize of p^n	Cost in core-years	Algorithm	Work
2017	$\mathbb{F}_{p^6}$	422	26.1	NFS	[18]
2020	$\mathbb{F}_{p^6}$	423	9.3	NFS	[29]
2021	$\mathbb{F}_{p^6}$	521	2.8	TNFS	[12]
2022	$\mathbb{F}_{p^4}$	512	6.3	TNFS	[32]

Table 1: Cost in core-years of the last discrete logarithm records on finite fields with composite extension degrees.

NFS and all its variants are built around four main steps, *polynomial selection*, *relation collection*, *linear algebra*, and *individual logarithm*. Asymptotically speaking, the *relation collection* and the *linear algebra* steps are equally hard and are the two most costly steps in all NFS variants. Experimental data from discrete logarithm records shows that they are the two most costly steps in practice as well. Table 2 shows the cost of the relation collection step, the linear algebra step and the whole computation of recent records. Although the relation collection have a higher cost than the linear algebra in these records except one, It is worth emphasizing that relation collection parallelizes much better than the linear-algebra step. Additionally, adjusting some parameters could balance the costs of these steps, and this strategy was followed to make the linear algebra step feasible in the 795-bit record [9].

Composite extension degrees in cryptographic applications. Pairing-based cryptography, introduced in the early 2000s [7,8,22], now underpins a range of applications—for example, the ECDAA protocol in the TPM2.0 Library [36]. Its security reduces to the hardness of the discrete logarithm problem in finite fields of extension degree $n > 1$ [30]. For efficiency, only a few pairing-friendly curves are used, constraining the underlying fields; in practice, composite extension degrees are the most deployed, such as $\mathbb{F}_{p^6}$ (MNT6 curves) and $\mathbb{F}_{p^{12}}$ (BLS12 curves). Interestingly, there is no good post-quantum candidates to replace pairing-based protocols yet.

Use of Galois automorphisms to accelerate the two hardest steps in NFS. For finite fields with extension degree $n > 1$, Galois symmetries can speed up NFS and its variants. Three subproblems arise: (i) compute some *NFS-compatible*

Bitsize of p^n	Finite Field	Cost in core days: Relation collection	Linear algebra	Total	Work
203	$\mathbb{F}_{p^{12}}$	10.5	0.28	11	[21]
324	$\mathbb{F}_{p^5}$	359	11.5	386	[17]
512	$\mathbb{F}_{p^4}$	878	20	368	[32]
521	$\mathbb{F}_{p^6}$	388	23	413	[12]
593	$\mathbb{F}_{p^3}$	3287	5113	8400	[14]
595	$\mathbb{F}_{p^2}$	157	18	175	[4]
795	$\mathbb{F}_p$	$876,000$	$228,125$	$116,800,0$	[9]

Table 2: Costs of the relation collection, the linear algebra and the whole computation in the last discrete logarithm records with various extension degrees.

automorphisms of order k dividing n; (ii) use them to accelerate the relation collection by about a factor k; and (iii) use them to shrink the linear system in the linear algebra step by a factor k. Because the complexity of this step is nearly quadratic due to the sparsity of the linear system, solving (iii) would yield an acceleration factor for the linear algebra step as large as k^2. In [4, 40] the authors provide polynomial selection methods to construct *NFS-compatible* automorphisms for orders $k = 2, 3, 4$ and 6, and problem (i) remains open for other orders. The solution to (ii) is straightforward. However, problem (iii) is only solved for $k = 2$ [4, 13, 40]. It was put into practice in the last discrete logarithm record with TNFS on $\mathbb{F}_{p^6}$ [13] which allowed to accelerate the linear algebra step by approximately a factor of 4.

Our work. We solve (iii) for $k = 6$, and 12. Specifically, our work focuses on accelerating the linear algebra step in the TNFS algorithm when applied to finite fields with extension degrees 6 and 12. Given any finite field of extension degree 6, resp. 12, we first show using the work in [4] a method to construct *NFS-compatible* automorphisms of order 6, resp. 12. Second, we present a new construction that allows the use of these automorphisms to accelerate the linear algebra step in TNFS with a factor approximately equals to 36, resp. 144. Third, we provide an implementation of TNFS in SageMath together with our construction to illustrate our findings on small size finite fields

Outline of the article. We start with a description of TNFS in Section 2. Section 3 defines the Galois automorphisms that are useful in this work. Section 4 exposes the conditions and obstacles and reviews the literature on using automorphisms to accelerate the linear algebra step. In Section 5 we present a new construction and we show in Section 6 that the new construction allows to accelerate the linear algebra step with factors of 36 and 144 in finite fields of extension degrees 6 and 12 respectively. Finally, Section 7 presents our experiments that validate our findings.

2 Background

2.1 The Tower Number Field Sieve

We target a finite field $\mathbb{F}_{p^n}$ where n is composite. Let η be a non-trivial divisor of n and denote $\kappa = n/\eta$. Since the computation of a discrete logarithm in a group can be reduced to its computation in one of its prime subgroups by Pohlig-Hellman's reduction, we will search for discrete logarithms modulo ℓ, a non trivial prime divisor of $\Phi_n(p)$, with Φ_n the n-th cyclotomic polynomial. The classical TNFS setup considers the intermediate number field $\mathcal{K}_h = \mathbb{Q}(\iota)$ where ι is a root of h, a polynomial of degree η over $\mathbb{Z}$ that remains irreducible modulo p. For a number field $\mathcal{K}$, we let $\mathcal{O}_{\mathcal{K}}$ be its ring of integers. For simplicity, we assume throughout this article that $\mathcal{O}_{\mathcal{K}_h} = \mathbb{Z}[\iota]$. This implies that h is monic.

Above $\mathcal{K}_h$, define two number fields $\mathcal{K}_1 = \mathcal{K}_h[x]/f_1(x)$ and $\mathcal{K}_2 = \mathcal{K}_h[x]/f_2(x)$ where f_1, f_2 are irreducible polynomials over $\mathcal{O}_{\mathcal{K}_h}$ that share an irreducible factor φ of degree κ modulo the unique ideal $\mathfrak{p}$ over p in $\mathcal{K}_h$. In particular, f_1 and f_2 have degree at least κ. Let α_i be root of f_i in $\mathcal{K}_i$ for $i = 1, 2$. Because of the conditions on the polynomials h, f_1 and f_2, there exist two ring homomorphisms from $\mathcal{O}_{\mathcal{K}_h}[x]$ to the target finite field $\mathbb{F}_{p^n}$ through the number fields $\mathcal{K}_1$ and $\mathcal{K}_2$. This allows to build a commutative diagram as in Figure 3. When η and κ are coprime (which is always possible with $n = 6$ or $n = 12$), then f_1 and f_2 can be defined over $\mathbb{Z}$. We make this choice for the rest of the article.

$$
\begin{tikzcd}
 & \mathcal{O}_{\mathcal{K}_h}[x] \arrow[dl] \arrow[dr] & \\
\mathcal{K}_1 \supset \mathcal{O}_{\mathcal{K}_h}[x]/f_1(x) \arrow[dr, "\Psi_1 = \bmod \varphi,\ \bmod p"'] & & \mathcal{K}_2 \supset \mathcal{O}_{\mathcal{K}_h}[x]/f_2(x) \arrow[dl, "\Psi_2 = \bmod \varphi,\ \bmod p"] \\
 & \mathcal{O}_{\mathcal{K}_h}/\mathfrak{p}[x]/\varphi(x) \cong \mathbb{F}_{p^n} &
\end{tikzcd}
$$

Diagram 3: Commutative diagram of Tower NFS.

The general framework of TNFS is common to all its variants. The *polynomial selection* step sets up the commutative diagram of Figure 3 by selecting adequate polynomials h, f_1 and f_2. Thereafter, by finding many small to medium size smooth numbers and factoring them, the *relation collection* step establishes linear relations involving prime ideals of small norms in the number fields $\mathcal{K}_1$ and $\mathcal{K}_2$. The unknowns of these equations are the values on the small prime ideals of a linear map called a *virtual logarithm map*. When enough equations are found, the *linear algebra* step solves the system. Given a target T in the finite field, the last step called the *individual logarithm* step establishes a linear equation between the logarithm of T and the virtual logarithms of the small ideals, which reveals the logarithm of the target.

Polynomial selection. Several methods to do TNFS polynomial selection are known. For example, the Conjugation, JLSV or Sarkar-Singh's methods [4, 23, 33] can be used. Each polynomial selection method yields different degrees and coefficient sizes for h, f_1 and f_2 which influence the whole performance of the algorithm. Based on the recent records [12, 32], the Conjugation method seems to perform best in practice, and the work in [4] shows how to construct adequate automorphisms using this method. Therefore, it is the only polynomial selection method considered in this work.

Conjugation polynomial selection [4]. First a polynomial h of degree η and with small integer coefficients that remains irreducible modulo p is chosen. Second, a quadratic irreducible polynomial μ over $\mathbb{Z}$ with small coefficients is initially selected, which possesses a root λ modulo p. Third, two polynomials g_0 and g_1 with small integer coefficients are chosen with the condition that $\deg(g_1) < \deg(g_0) = \kappa$ and $\varphi := g_0 + \lambda g_1$ is irreducible modulo p. The polynomial f_1 is defined as $f_1 := \mathrm{Res}_Y(\mu(Y), g_0 + Y g_1)$, and f_2 is defined as $f_2 := v g_0 + u g_1$ where $\frac{u}{v} \equiv \lambda \mod p$ is a rational reconstruction of λ. Given that $f_1 \equiv 0 \mod p \mod \varphi$ and $f_2 \equiv v\varphi \mod p$, both share φ as an irreducible factor modulo p. Their respective degrees are 2κ and κ.

Relation collection. The goal of the relation collection step is to select, among the set of polynomials $\phi(x, \iota) \in \mathcal{O}_{\mathcal{K}_h}[x]$ at the top of Diagram 3, the candidates that yield a relation. A relation is found if the polynomial $\phi(x, \iota)$ maps to principal ideals in $\mathcal{O}_{\mathcal{K}_1}$ and $\mathcal{O}_{\mathcal{K}_2}$ that are *smooth* (respectively B_1- and B_2-smooth for some bounds B_1 and B_2). Most often the search space for relation collection consists of linear polynomials $\phi(x, \iota) = a(\iota) - b(\iota)x \in \mathcal{O}_{\mathcal{K}_h}[x]$. The ideals that occur in the factorizations in $\mathcal{O}_{\mathcal{K}_1}$ and $\mathcal{O}_{\mathcal{K}_2}$ constitute the factor basis $\mathcal{F}$. More precisely, we define it as the disjoint union $\mathcal{F} = \mathcal{F}_1 \sqcup \mathcal{F}_2$ with, for $i = 1, 2$:

$$\mathcal{F}_i(B_i) = \{\text{prime ideals of } \mathcal{O}_{\mathcal{K}_i} \text{ of norm} \le B_i \text{ and inertia degree 1 over } \mathcal{K}_h\}.$$

An ideal $(a(\iota) - b(\iota)\alpha_i)$ is B_i-smooth if and only if its norm $\mathcal{N}_i(a(\iota) - b(\iota)\alpha_i)$ is B_i-smooth for $i = 1, 2$, where $\mathcal{N}_i$ defines the algebraic norm in $\mathcal{K}_i$. Working with norms, which are integer values, enables efficient sieving algorithms for this step. The relation collection stops when we have enough relations to construct a system of linear equations that may be full rank. The unknowns of these equations are *virtual logarithms* of the ideals of the factor basis. A virtual logarithm map is a linear map with values in $Z/\ell\mathbb{Z}$. Its definition is detailed in Section 2.2.

Linear algebra. A good feature of the linear system created is that the number of non-zero coefficients per row is very small. This allows to use sparse linear algebra algorithms in $\mathbb{Z}/\ell\mathbb{Z}$ such as the Wiedemann algorithm [39] or Coppersmith's block Wiedemann algorithm [10], which allows partial parallelization. The output of this step is a kernel vector corresponding to virtual logarithms of the ideals in the factor basis.

Individual discrete logarithm. The final step consists in finding the discrete logarithm of one or several target elements. This step is subdivided into two substeps: a smoothing step and a descent step. The smoothing step is an iterative process where the target element is randomized until the randomized value lifted back to one of the number fields $\mathcal{K}_i$ is B_i'-smooth for a smoothness bound $B_i' > B_i$, for $i = 1$ or 2. The second step consists in decomposing every factor of the lifted value, in our case prime ideals with norms less than a smoothness bound B_i', into elements of the factor basis for which we now know the virtual logarithms. This eventually makes it possible to reconstruct the discrete logarithm of the target element. TNFS differs from NFS in this step as there exist improvements for the smoothing step when the target finite field has proper subfields [2, 20].

2.2 Schirokauer and virtual logarithm maps

This Section follows the presentation in [38]. We drop the subscript and consider $\mathcal{K}$ one of the two number fields. Define

$$\Gamma := \{\phi \in \mathcal{K}^* \mid (\phi) \text{ factors in the factor basis}\}.$$

Since two generators of an ideal differ by a unit, to properly represent an element $\gamma \in \Gamma$, we need both its ideal factorization and some canonical choice of a unit associated to γ. Let $\mathcal{O}^\times$ denote the unit group of $\mathcal{K}$, i.e, the invertible elements of $\mathcal{O}$. Dirichlet's unit theorem provides the following abelian group isomorphism

$$\mathcal{O}^\times \simeq \mu_{\mathcal{K}} \times \mathbb{Z}^r,$$

where $\mu_{\mathcal{K}}$ is the group of roots of unity of $\mathcal{K}$ and r is an integer called the *unit rank* of $\mathcal{K}$. A $\mathbb{Z}$-basis of the non-torsion part (i.e, $\mathbb{Z}^r$) is called a *system of fundamental units.*

Proposition 1 (Structure of Γ/Γ^ℓ). *Let $\mathcal{F}$ the factor basis on side $\mathcal{K}$, $\mathcal{I}$ the group of fractional ideals that completely factor in $\mathcal{F}$, Γ and $\mu_{\mathcal{K}}$ defined as above, and recall that ℓ is a prime divisor of $\Phi_n(p)$. Then, Γ/Γ^ℓ is a finite dimensional $\mathbb{F}_\ell$-vector space. Further, suppose that the class number of $\mathcal{K}$ denoted as $h_{\mathcal{K}}$ and $|\mu_{\mathcal{K}}|$ are both coprime to ℓ, then we have the following vector space isomorphism*

$$\Gamma/\Gamma^\ell \simeq \mathcal{O}^\times/(\mathcal{O}^\times)^\ell \times \mathcal{I}/\mathcal{I}^\ell,$$

with basis $\{u_1, \ldots, u_r\} \cup \mathcal{F}/\mathcal{F}^\ell$, where, $\{u_1, \ldots, u_r\}$ is a system of fundamental units of $\mathcal{K}$ considered modulo $(\mathcal{O}^\times)^\ell$. In particular, $\dim(\Gamma/\Gamma^\ell) = r + |\mathcal{F}|$.

Proof. We found no proof of this in the literature. We sketch one in Appendix A.1.

For every number field $\mathcal{K}$, we always suppose its discriminant $\operatorname{disc}(\mathcal{K})$, $h_{\mathcal{K}}$, r, and $\mu_{\mathcal{K}}$ all coprime to ℓ. This occurs with very large propability as ℓ is large.

Schirokauer maps By Proposition 1, fundamental units (with ideal factorization) would give explicit coordinates for Γ/Γ^ℓ, but computing them is computationally hard. Schirokauer maps [35] circumvent this, under an additional hypothesis.

Definition 1 (Usual definition of a Schirokauer map). *Consider Γ from Proposition 1 and denote r the unit rank of $\mathcal{K}$. A Schirokauer map on Γ/Γ^ℓ is any full-rank $\mathbb{F}_\ell$-linear map $\Gamma/\Gamma^\ell \longrightarrow (\mathbb{Z}/\ell\mathbb{Z})^r$ that still has full rank when restricted to $\mathcal{O}^\times/(\mathcal{O}^\times)^\ell$.*

In other words, when restricted to $\mathcal{O}^\times/(\mathcal{O}^\times)^\ell$ a Schirokauer map on $\mathcal{K}$ is a dual basis of a system of fundamental units.

Construction of a Schirokauer map. Write $\mathcal{K} = \mathbb{Q}[X]/(F)$ with $F \in \mathbb{Q}[X]$ irreducible and $\operatorname{disc}(F)$ coprime to ℓ. Consider the decomposition of F modulo ℓ into irreducible distinct factors: $F = \prod_j F_j \mod \ell$. By the Chinese remainder theorem we have the ring isomorphisms:

$$\mathcal{O}/\ell\mathcal{O} \simeq \mathbb{F}_\ell[X]/(F) \simeq \prod_j \mathbb{F}_\ell[X]/(F_j).$$

Define $\omega := \operatorname{lcm}\left(\ell^{\deg(F_j)} - 1\right)$ and consider $x \in \Gamma \cap \mathcal{O}$. By the above isomorphisms we have $x^\omega \equiv 1 \mod \ell\mathcal{O}$. Further, lifting the congruence modulo $\ell^2\mathcal{O}$ we get $x^\omega \equiv 1 + \ell \cdot P(x) \mod \ell^2\mathcal{O}$ for some $P \in \mathbb{Z}[X]$ with degree bounded by $\deg(F) - 1$. We get the following linear map

$$\mathbf{A} : \begin{cases} \Gamma/\Gamma^\ell \longrightarrow & \mathbb{F}_\ell[X]/(F) \\ \quad x \quad \longmapsto & \frac{x^\omega - 1}{\ell} \mod \ell\mathcal{O} := P(x) \end{cases} \tag{1}$$

As far as we know, all practically computable Schirokauer map constructions (including ours presented in later sections) are based on the map $\mathbf{A}$.

The Schirokauer map used in the literature $\Lambda : \Gamma/\Gamma^\ell \to (\mathbb{Z}/\ell\mathbb{Z})^r$ is *defined* over an element x as the first r (or any r independent integer linear combinations of the) coefficients of $\mathbf{A}(x)$. This provides a linear map that is computable in polynomial time. We emphasize that there is no efficient way for testing whether it is a Schirokauer map or not in that it preserves the rank, in particular because there is no efficient way for computing a system of fundamental units. Consequently, we need to rely on the following usual assumption.

Assumption 1 *Λ has maximal rank on $\mathcal{O}^\times/(\mathcal{O}^\times)^\ell$.*

Denote $\{\mathfrak{p}_1, \ldots, \mathfrak{p}_b\}$ the factor basis on side $\mathcal{K}$, then under Assumption 1, every $\phi \in \Gamma/\Gamma^\ell$ is represented as

$$\phi \leftrightarrow (\Lambda(\phi)_1, \ldots, \Lambda(\phi)_r, \operatorname{val}_{\mathfrak{p}_1}(\phi), \ldots, \operatorname{val}_{\mathfrak{p}_b}(\phi)) \tag{2}$$

In other words, $(\Lambda(\cdot)_i) \sqcup (\operatorname{val}_{\mathfrak{p}_i}(\cdot))_i$ form a generating family of the dual $(\Gamma/\Gamma^\ell)^\dagger$.

Virtual logarithm maps Consider both $\mathcal{K}_1$ and $\mathcal{K}_2$ from Figure 3 with Γ_i and a Schirokauer map Λ_i on each side.

Definition 2 (Virtual logarithm). *A virtual logarithm is a pair of $\mathbb{F}_\ell$-linear forms $(\mathrm{vlog}_1, \mathrm{vlog}_2)$ where for $i = 1, 2$ $\mathrm{vlog}_i : \Gamma_i/\Gamma_i^\ell \longrightarrow \mathbb{Z}/\ell\mathbb{Z}$ and such that for all $\phi \in \mathcal{O}_{\mathcal{K}_h}[x]$ that projects to both $\phi_1 \in \Gamma_1$ and $\phi_2 \in \Gamma_2$ we have*

$$\mathrm{vlog}_1(\phi_1) = \mathrm{vlog}_2(\phi_2).$$

The goal of the relation collection and the linear algebra is to compute a virtual logarithm map. Under Assumption 1 on Λ_1 and Λ_2, denote $\{u_{j,i}\}_{i=1}^{r_j}$ for $j = 1, 2$ for the dual basis of Λ_j, which is *not computed*. When an element ϕ is smooth on both sides, with virtual logarithms as unknowns.

$$\begin{aligned} &\sum_{i=1}^{r_1} \Lambda(\phi_1)_i \mathrm{vlog}(u_{1,i}) + \sum_{i=1}^{b_1} \mathrm{val}_{\mathfrak{p}_{1,i}}(\phi_1) \mathrm{vlog}(\mathfrak{p}_{1,i}) \\ =&\sum_{i=1}^{r_2} \Lambda(\phi_2)_i \mathrm{vlog}(u_{2,i}) + \sum_{i=1}^{b_2} \mathrm{val}_{\mathfrak{p}_{2,i}}(\phi_2) \mathrm{vlog}(\mathfrak{p}_{2,i}), \end{aligned} \tag{3}$$

3 TNFS Automorphisms

The following is a definition of automorphisms that are useful in TNFS. It extends the definition provided in [12].

Definition 3 (TNFS Automorphism). *Consider $\mathbb{F}_{p^n}$ and the absolute fields $\mathcal{K}_1$ and $\mathcal{K}_2$ (i.e, defined over $\mathbb{Q}$) from Figure 3. A TNFS automorphism is a couple $(\sigma_1, \sigma_2) \in \mathrm{Aut}(\mathcal{K}_1) \times \mathrm{Aut}(\mathcal{K}_2)$ fullfilling two conditions:*

- ***Both are Frobenius.*** *Each fix a degree n prime ideal $\mathfrak{p}_i$ over p in $\mathcal{O}_{\mathcal{K}_i}$.*
- ***Project to the same Frobenius.*** *There exists $\overline{\sigma} \in \mathrm{Gal}(\mathbb{F}_{p^n}/\mathbb{F}_p)$ such that for all $(\phi_1, \phi_2) \in \mathcal{O}_h[x]/(f_1) \times \mathcal{O}_h[x]/(f_2)$:*

  $$\Psi_1(\sigma_1(\phi_1)) = \overline{\sigma}(\Psi_1(\phi_1)) \quad \textit{and} \quad \Psi_2(\sigma_2(\phi_2)) = \overline{\sigma}(\Psi_2(\phi_2)),$$

  where (Ψ_1, Ψ_2) are the projection morphisms from Figure 3.

We emphasize that the restrictive condition is the "both are Frobenius". The other condition follows by raising σ_1 and σ_2 to right powers.

Construction of Galois automorphisms with the Conjugation polynomial selection. To construct degree n automorphisms in both $\mathcal{K}_1$ and $\mathcal{K}_2$, it is sufficient to choose the three polynomials h, f_1, and f_2 such that $\mathcal{K}_h$ has a degree η automorphism that we term as σ_h and each of $\mathcal{K}_{f_i} := \mathbb{Q}[x]/(f_i)$ has a degree κ automorphism denoted as σ_{f_i} for $i = 1, 2$. Indeed, since η and κ are coprime, for $i = 1, 2$, the automorphism σ_i of $\mathcal{K}_i$ defined by the joint action of σ_h and σ_{f_i} has order $\eta \times \kappa = n$. In [4], the authors provide choices of g_0 and g_1 for the Conjugation polynomial selection presented in §2.1 that provide automorphisms of $\mathcal{K}_{f_1}$ and $\mathcal{K}_{f_2}$ of orders equal to 2, 3, 4 and 6. Table 4 presents these choices.

Degree of $(\mathcal{K}_{f_1}, \mathcal{K}_{f_2})$	Automorphism's order	g_0	g_1 $(a \in \mathbb{Z}^*)$	Automorphism $\alpha \mapsto$
$(4,2)$	2	x^2+1	ax	$1/\alpha$
		x^2-1	ax	$-1/\alpha$
		x^2	a	$-\alpha$
$(6,3)$	3	x^3-3x-1	$-a(x^2+x)$	$-(\alpha+1)/\alpha$
$(8,4)$	4	x^4-6x^2+1	$a(x^3-x)$	$-(\alpha+1)/(\alpha-1)$
$(12,6)$	6	x^6+6x^5- $20x^3-15x^2+1$	$a(2x^5+5x^4-$ $5x^2-2x)$	$-(2\alpha+1)/(\alpha-1)$

Table 4: Table from [4]. Automorphisms with the Conjugation method §2.1. The rational expression of the automorphism and its order are the same for both number fields. The automorphism is also an automorphism of g_0+g_1.

4 Speed-up with TNFS automorphisms

A TNFS automorphism of order k allows to accelerate the relation collection step by approximately a factor of k. This is due to the fact that the factor basis is stable under the automorphisms action since the conjugate of a prime ideal is a prime ideal of equal norm. Consequently, conjugating an equation on both sides provides "for free" a new equation, and iterating with powers of the automorphisms yields the claimed speed-up factor.

As for the linear algebra step, one hopes that the virtual logarithms of k conjugate ideals can be recovered from the virtual logarithm of one of these ideals. This would reduce the matrix size by a factor of k and hence accelerate the linear algebra step by approximately a factor of k^2 since its complexity is nearly quadratic. Unfortunately, this is not true in general. The reason of failure is due to complications related to Schirokauer maps and the units, as we examine in the subsequent.

Notations. Let $\mathcal{K}$ one of the middle fields from Figure 3, h its class number, $\{\mathfrak{p}_i\}_{i=1}^b$ a factor basis, Λ a Schirokauer map, and σ an order k TNFS automorphism, all on side $\mathcal{K}$. Denote $\overline{\sigma} \in \mathrm{Gal}(\mathbb{F}_{p^n}/\mathbb{F}_p)$ the corresponding Frobenius which expresses as $\overline{\sigma} : x \mapsto x^\zeta$, where ζ is a k-th primitive root of unity modulo ℓ. Notations $\mathcal{K}^{\sigma^s}$ and $\mathbb{F}_{p^n}^{\overline{\sigma}^s}$ refer to the subfield fixed point-wise by σ^s and $\overline{\sigma}^s$ respectively, with s an integer. Additionally, under Assumption 1 on Λ, let $\mathbf{u} := \{u_i\}_{i=1}^r$ denotes its dual basis (thus a system of fundamental units), where r is the unit rank. The notation $\mathrm{vlog}(\mathbf{u})$ refers to the r dimensional vector $(\mathrm{vlog}(u_1), \ldots, \mathrm{vlog}(u_r))$. If ϕ belongs to Γ defined in Section 2.2, we abusively drop $\bmod \Gamma^\ell$ and write $\mathrm{vlog}(\phi)$ and $\Lambda(\phi)$. Last but not least, recall that $\mathrm{disc}(K)$, r, h, and $|\mu_{\mathcal{K}}|$ are coprime with ℓ.

Remark 1. We have for all $x \in \mathbb{F}_{p^n}^\times$, $\log\left(x^{\overline{\sigma}}\right) \equiv \zeta \log(x) \bmod \ell$. Our goal is to construct a virtual logarithm map that verifies $\mathrm{vlog}(\mathfrak{p}^\sigma) \equiv \zeta\,\mathrm{vlog}(\mathfrak{p}) \bmod \ell$ for all prime ideals $\mathfrak{p}$ in the factor basis, which would shrink our linear system.

4.1 Obstacles to speed-up the linear algebra step

A virtual logarithm vanishes on the elements fixed by a TNFS automorphism.

Lemma 1. *Let $\phi \in \Gamma$ such that $\sigma(\phi) = \phi$, then* $\mathrm{vlog}(\phi) = 0$.

Proof. There exists $\tilde{g}$ a generator of $\mathbb{F}_{p^n}^{\times}$ such that $\mathrm{vlog}(\phi) = \log_{\tilde{g}}(\overline{\phi}) \bmod \ell$, where $\overline{\phi}$ is the projection of ϕ to $\mathbb{F}_{p^n}$. by Definition 3 we have $\overline{\sigma}(\overline{\phi}) = \overline{\sigma(\phi)} = \overline{\phi}$, hence, $\overline{\phi} \in \mathbb{F}_{p^n}^{\overline{\sigma}}$. Lemma 1 from [19] states that the logarithm of a proper subfield element vanishes modulo ℓ, which yields $\log_{\tilde{g}}(\overline{\phi}) \equiv 0 \mod \ell$.

Applying the lemma to powers of σ, we get the following corollary.

Corollary 1. *Let $\phi \in \Gamma$ such that there exists s a proper divisor of k with $\sigma^s(\phi) = \phi$. Then* $\mathrm{vlog}(\phi) = 0$.

The above lemma yields a link between the virtual logarithms of conjugate ideals.

Proposition 2. *Let $\mathfrak{p}$ a prime ideal. Then there exists $a \in \mathcal{K}^{\sigma}$ such that*

$$\sum_{i=0}^{k-1} \mathrm{vlog}\left(\mathfrak{p}^{\sigma^i}\right) = -h^{-1}\ \Lambda(a) \cdot \mathrm{vlog}(\mathbf{u}).$$

Proof. By the class group theorem, there exists $\gamma \in \mathcal{K}$ such that $(\gamma) = \mathfrak{p}^h$. Therefore, for all $i \in [\![0, k-1]\!]$, $(\gamma^{\sigma^i}) = (\mathfrak{p}^{\sigma^i})^h$. Applying the virtual logarithm map we get for all $0 \le i \le k-1$, $\mathrm{vlog}\left(\gamma^{\sigma^i}\right) = \Lambda\left(\gamma^{\sigma^i}\right) \cdot \mathrm{vlog}(\mathbf{u}) + h\ \mathrm{vlog}\left(\mathfrak{p}^{\sigma^i}\right)$. Summing the k equations we get

$$\mathrm{vlog}\left(\prod_{i=0}^{k-1} \gamma^{\sigma^i}\right) = \Lambda\left(\prod_{i=0}^{k-1} \gamma^{\sigma^i}\right) \cdot \mathrm{vlog}(\mathbf{u}) + h \sum_{i=0}^{k-1} \mathrm{vlog}\left(\mathfrak{p}^{\sigma^i}\right).$$

The left hand side is zero by Lemma 1 and the result follows with $a = \prod_{i=0}^{k-1} \gamma^{\sigma^i}$.

Imposing $\mathrm{vlog}(\mathfrak{p}^{\sigma}) = \zeta\,\mathrm{vlog}(\mathfrak{p})$ forces the sum in Proposition 2 to be zero, so the right-hand side of the equality vanishes as well. Since computing h, a, or $\mathbf{u}$ is hard, we need careful virtual logarithm constructions to ensure this vanishing without the need to compute these quantities. The property we need is formalized by the following condition.

Condition 1 *For each prime ideal $\mathfrak{p}$ in the factor basis,*

$$\sum_{i=0}^{k-1} \mathrm{vlog}\left(\mathfrak{p}^{\sigma^i}\right) = 0.$$

In vlog constructions verifying Condition 1, we may replace every k conjugate unknowns $\mathrm{vlog}(\mathfrak{p})$, $\mathrm{vlog}(\mathfrak{p}^{\sigma})$, …, $\mathrm{vlog}(\mathfrak{p}^{\sigma^{k-1}})$ by one unknown $\mathrm{vlog}(\mathfrak{p})$, $\zeta\,\mathrm{vlog}(\mathfrak{p})$, …, $\zeta^{k+1}\,\mathrm{vlog}(\mathfrak{p})$, thus shrinking the system size by k and forcing the linear algebra step to compute a virtual logarithm map that equates conjugate ideals up to multiplication by a primitive root of unity. More details about the shrinking and the speed-up factors are discussed in Section 5.3. We emphasize that without Condition 1, the above constraints would make the linear system incompatible.

4.2 Literature on speeding up the linear algebra step

In this section we examine constructions from the literature in which Condition 1 is satisfied. All these constructions work only with order 2 TNFS automorphisms.

Vanishing virtual logarithm on units for order two automorphisms A *complex multiplication field* (CM field) $\mathcal{K}$ is a number field that is totally imaginary and that has a totally real subfield of index 2. Let $\mathcal{K}$ from the middle of Figure 3 be CM, and σ an order k TNFS automorphism on side $\mathcal{K}$. We say that $\mathcal{K}$ is *CM TNFS (relatively to σ)* if its real subfield of index two is $\tilde{\mathcal{K}} := \mathcal{K}^{\sigma^{k/2}}$.

Proposition 3. *Let $\mathcal{K}$ CM TNFS with σ of order k and $\tilde{\mathcal{K}} := \mathcal{K}^{\sigma^{k/2}}$ its real subfield of index 2. Then, the index $v := [\mathcal{O}_{\mathcal{K}}^{\times} : \mathcal{O}_{\tilde{\mathcal{K}}}^{\times}]$ is finite and if v is coprime to ℓ, then* $\mathrm{vlog}(u) = 0$ *for all $u \in \mathcal{O}_{\mathcal{K}}^{\times}/(\mathcal{O}_{\mathcal{K}}^{\times})^{\ell}$.*

Proof. For completeness, we sketch a proof in Appendix A.2.

Note that v will be coprime to ℓ with very large probability. The above proposition and Proposition 2 imply that Condition 1 is satisfied. In fact, if both fields $\mathcal{K}_1$ and $\mathcal{K}_2$ from Figure 3 are CM TNFS, then the virtual logarithm vanishes on all the units and no Shirokauer map is needed. Nevertheless, requiring such constructions is very restrictive and the only known methods to do so are with order 2 automorphisms.

In [4], Barbulescu et al. construct TNFS diagrams with double-sided CM fields for extension degrees 4 and 6, but only with TNFS automorphisms of order 2. Zhu et al. [40] generalize this to arbitrary even extension degree finite fields. However, to our knowledge, there are no known constructions for double sided CM fields with an automorphism of order larger than 2.

Partial vanishing over the units. In the slides "How to get rid of units"[2], Barbulescu proposes alternative constructions where the virtual logarithm map vanishes on only a subset of the units. The examples use number fields $\mathcal{K}$ with automorphisms σ of orders 2, 3, 4, and 6, where $\mathcal{K}^{\sigma}$ forces some units to have zero virtual logarithm. Our work is an extension of this idea. We explore the full Galois treilli $\{\mathcal{K}^{\sigma^i}\}$ and introduce adequate *Galois Schirokauer map.*

Vanishing Schirokauer map for order two automorphisms By Proposition 2, the other alternative to ensure Condition 1 is to construct a Schirokauer map that vanishes on the field elements fixed by the automorphism. In the 521-bit record [13] on $\mathbb{F}_{p^6}$, De Micheli et al. construct such a Schirokauer map with an order 2 TNFS-automorphism σ. We now examine their construction.

Since $\mathcal{K}^{\sigma}$ has index 2, let $\omega \in \mathcal{K}$ such that $\mathcal{K} = \mathcal{K}^{\sigma}(\omega)$. For $\gamma \in \Gamma/\Gamma^{\ell}$, compute $\mathbf{A}(\gamma)$ from (1) and decompose it as $\mathbf{A}(\gamma) = \gamma_0 + \gamma_1\omega \bmod \ell\mathcal{O}$ with γ_0 and γ_1

[2] `https://www.lix.polytechnique.fr/~guillevic/catrel-workshop/Razvan_Barbulescu_CATRELworkshop.pdf`

having polynomial representation over $\mathbb{F}_\ell$ of degree $m/2-1$, where m denotes the degree of $\mathcal{K}$ over $\mathbb{Q}$. A Schirokauer map Λ on γ is defined by taking r independent integer linear combinations of the coefficients of γ_1, which is possible if and only if $r \leq m/2$. This Schirokauer map vanishes on the elements fixed by σ. Indeed, if γ belongs to $\mathcal{K}^\sigma$, then so does any lift of $\mathbf{A}(\gamma)$, and hence γ_1 is zero and so is $\Lambda(\gamma)$. The authors provide two such constructions on two finite field instances of degrees 6 and 12. However, the construction is specific to order 2 automorphisms. Generalizing the construction to order k automorphisms requires $r \leq m/k$, which is too restrictive. Specifically, since $r \geq m/2+1$ by Dirichlet's theorem, the above requirement fails whenever $k \geq 4$ or ($k=3$ and $m>6$).

5 New construction: Galois Schirokauer map

5.1 Construction

We propose a new Schirokauer map construction that not only vanishes on the elements fixed by the automorphism σ, but also on the ones fixed by σ^s where s is a proper divisor of σ's order k (including 1). It comes at the cost of reducing the rank of the Schirokauer map. The following extends the definition from [13].

Definition 4 (Galois Schirokauer map (GSM)). *A Galois Schirokauer map (GSM) on $\mathcal{K}$ is any full rank linear map $\Lambda : \Gamma/\Gamma^\ell \longrightarrow (\mathbb{Z}/\ell\mathbb{Z})^w$ for some integer w that still has full rank when restricted to $\mathcal{O}^\times/(\mathcal{O}^\times)^\ell$ and such that $\Lambda(\phi)=0$ whenever there exists s a proper divisor of k with $\sigma^s(\phi)=\phi$. The integer w is referred to as Λ's dimension.*

The rank w of a GSM will be smaller than the unit rank r of the field (as opposed to the usual Definition 1). Nevertheless, we will prove that the property of vanishing on $\{\mathcal{K}^{\sigma^s}\}_{s|k, s\neq k}$ is enough compensation in many cases.

Construction. The TNFS automorphism's order k divides m the degree of $\mathcal{K}$ over $\mathbb{Q}$. Let $\omega := m/k$ and d a divisor of k (to be determined later). Recall that $\zeta \in \mathbb{F}_\ell$ is the k-th primitive root of unity such that $\overline{\sigma} : x \mapsto x^\zeta$. Further, we need a $\mathbb{Q}$-*Galois-compatible-basis* of $\mathcal{K}$. Let $\alpha_0, \ldots, \alpha_{\omega-1} \in \mathcal{K}^\sigma$ and $y \in \mathcal{K}$ such that $\{a_i y^{\sigma^j}\}_{0\leq i\leq \omega-1,\ 0\leq j\leq k-1}$ is a $\mathbb{Q}$-basis of $\mathcal{K}$. Our construction is the following.

Definition 5 (Construction of GSM). *$\Lambda_d : \Gamma/\Gamma^\ell \to (\mathbb{Z}/\ell\mathbb{Z})^{d\cdot\omega}$ is defined by:*

- *First, compute the image of γ by $\mathbf{A}$ defined in* (1).
- *Second, decompose $\mathbf{A}(\gamma)$ in the above basis as*

$$\mathbf{A}(\gamma) \equiv \sum_{i=0}^{\omega-1}\sum_{j=0}^{k-1} \gamma_{i,j}\ a_i y^{\sigma^j} \mod \ell\mathcal{O},$$

where for $0\leq i\leq \omega-1$ and $0\leq j\leq k-1$, the scalar $\gamma_{i,j}$ belongs to $\mathbb{F}_\ell$.

– *Third, Λ_d is defined on γ in matrix form by:*

$$\Lambda_d : \begin{cases} \Gamma/\Gamma^{\ell} \longrightarrow & (\mathbb{Z}/\ell\mathbb{Z})^{d\cdot\omega} \\ \quad\gamma \quad \longmapsto & \begin{pmatrix} \sum\limits_{\substack{j=0 \\ j\equiv 0[d]}}^{k-1} \gamma_{0,j}\ \zeta^j & \dots & \sum\limits_{\substack{j=0 \\ j\equiv 0[d]}}^{k-1} \gamma_{\omega-1,j}\ \zeta^j \\ \vdots & \dots & \vdots \\ \sum\limits_{\substack{j=0 \\ j\equiv d-1[d]}}^{k-1} \gamma_{0,j}\ \zeta^j & \dots & \sum\limits_{\substack{j=0 \\ j\equiv d-1[d]}}^{k-1} \gamma_{\omega-1,j}\ \zeta^j \end{pmatrix} \end{cases}.$$

The application Λ_d is a linear map. However, not any divisor d of k is convenient. It must be chosen to ensure that Λ_d vanishes on $\cup_{s|k,s\neq k}\mathcal{K}^{\sigma^s}$. The following lemma and corollary show the behavior of Λ_d with respect to σ and provide the largest possible d.

Lemma 2. *$\Lambda_d\left(\gamma^{\sigma}\right) = \zeta\mathbf{J}_d\Lambda_d(\gamma)$, where $\mathbf{J}_d$ is the following $d\times d$ circular matrix*

$$\mathbf{J}_d = \begin{pmatrix} 0 & 1 & & \\ & \ddots & \ddots & \\ & & 0 & 1 \\ 1 & & & 0 \end{pmatrix}.$$

Proof. Since σ fixes $\ell\mathcal{O}$, it is well defined on $\mathcal{O}/\ell\mathcal{O}$, and in particular on the image of $\mathbf{A}$. We abusively still refer to it as σ. First, since $\mathbf{A}(\gamma)$ is a polynomial expression in γ, then $\mathbf{A}(\gamma^{\sigma}) \equiv \mathbf{A}(\gamma)^{\sigma} \mod \ell\mathcal{O}$. Second, $\mathbf{A}(\gamma)^{\sigma}$ decomposes as:

$$\mathbf{A}(\gamma)^{\sigma} \equiv \sum_{i=0}^{\omega-1}\sum_{j=0}^{k-1}\gamma_{i,j}\ y^{\sigma^{j+1}} \equiv \sum_{i=0}^{\omega-1}\sum_{j=1}^{k}\gamma_{i,j-1}\ y^{\sigma^{j}} \mod \ell\mathcal{O},$$

Third, we have,

$$\begin{aligned} \Lambda_d\left(\gamma^{\sigma}\right) = \zeta &\begin{pmatrix} \sum\limits_{\substack{j=0 \\ j\equiv 0[d]}}^{k-1} \gamma_{0,j-1}\ \zeta^{j-1} & \dots & \sum\limits_{\substack{j=0 \\ j\equiv 0[d]}}^{k-1} \gamma_{\omega-1,j-1}\ \zeta^{j-1} \\ \vdots & \dots & \vdots \\ \sum\limits_{\substack{j=0 \\ j\equiv d-1[d]}}^{k-1} \gamma_{0,j-1}\ \zeta^{j-1} & \dots & \sum\limits_{\substack{j=0 \\ j\equiv d-1[d]}}^{k-1} \gamma_{\omega-1,j-1}\ \zeta^{j-1} \end{pmatrix} \\ = &\ \zeta\mathbf{J}_d\Lambda_d(\gamma). \end{aligned}$$

Corollary 2. *Let $d := \prod_{p,\mathrm{val}_k(p)>0} p^{\mathrm{val}_p(k)-1}$. Then d is the largest integer such that Λ_d vanishes on $\cup_{s|k,s\neq k}\mathcal{K}^{\sigma^s}$.*

Proof. Let $\tilde{d}$ a divisor of k. For e a proper divisor of k and $\gamma \in \mathcal{K}^{\sigma^e}$ we have by Lemma 2 $\Lambda_{\tilde{d}}(\gamma) = \Lambda_{\tilde{d}}\left(\gamma^{\sigma^e}\right) = \zeta^e\mathbf{J}_{\tilde{d}}^e\Lambda_{\tilde{d}}(\gamma)$. Thus, $\left(\zeta^e\mathbf{J}_{\tilde{d}}^e - \mathbf{I}_{\tilde{d}}\right)\Lambda_{\tilde{d}}(\gamma) = 0$.

We now examine at what condition on $\tilde{d}$ the matrices $\{\zeta^e \mathbf{J}^e_{\tilde{d}} - \mathbf{I}_{\tilde{d}}\}_{e|k, e\neq k}$ are all invertible. Given such an e, the matrix determinant is equal to $(-1)^{\tilde{d}}\zeta^{e\tilde{d}}\chi_{J^e_{\tilde{d}}}(\zeta^{-e})$, where $\chi_{J^e_{\tilde{d}}}$ is the characteristic polynomial of $J^e_{\tilde{d}}$. Since the eigenvalues of $J^e_{\tilde{d}}$ are $\{\zeta^{i\cdot e\cdot k/\tilde{d}}\}_{i=0}^{\tilde{d}-1}$, we have

$$\begin{aligned}
\chi_{J^e_{\tilde{d}}}(X) &= \prod_{i=0}^{\tilde{d}-1}\left(X - \zeta^{i\cdot e\cdot k/\tilde{d}}\right)\\
&= \left(\prod_{i=0}^{\frac{\tilde{d}}{\gcd(e,\tilde{d})}-1}\left(X - \zeta^{i\cdot e\cdot k/\tilde{d}}\right)\right)^{\gcd(e,\tilde{d})}\\
&= \left(X^{\frac{\tilde{d}}{\gcd(e,\tilde{d})}} - 1\right)^{\gcd(e,\tilde{d})},
\end{aligned}$$

where the second equality comes from $\min\{i \in \mathbb{N} \mid \zeta^{i\cdot e\cdot k/\tilde{d}} = 1\} = \min\{i \in \mathbb{N} \mid ie/\tilde{d} \in \mathbb{N}\} = \tilde{d}/\gcd(e,\tilde{d})$. Consequently, $\zeta^e\mathbf{J}^e_{\tilde{d}} - \mathbf{I}_{\tilde{d}}$ is invertible if and only if $\zeta^{-e\tilde{d}/\gcd(e\tilde{d})} \neq 1$, which is equivalent to $k \nmid \operatorname{lcm}(e,\tilde{d})$. Considering for e the values $\{k/p \mid \operatorname{val}_p(k) > 0\}$ shows that the announced d is the largest choice for $\tilde{d}$ that fulfills $k \nmid \operatorname{lcm}(e,\tilde{d})$ for all $s \mid k$, $s \neq k$.

With d from Corollary 2 and under Assumption 1 on Λ_d, its rank is equal to $d \cdot m/k$. Note that $d = 1$ if and only if k is square-free.

Example 1. Consider a TNFS diagram as in Figure 3 constructed with the conjugation polynomial selection with $\mathbb{F}_{p^6}$ (i.e, $n = 6$), $(\mathcal{K}_1, \mathcal{K}_2)$ of degree over $\mathbb{Q}$ equal to $(m_1 = 12, m_2 = 6)$, and a TNFS automorphism (σ_1, σ_2) of order 6 (i.e, $k = 6$). We now examine the GSM construction $(\Lambda_{d,1}, \Lambda_{d,2})$ on $(\mathcal{K}_1, \mathcal{K}_2)$.

- **GSM on side $\mathcal{K}_2$.** Since $k = 2$ is square-free, by Corollary 2 we have $d = 1$ and the rank of $\Lambda_{1,2}$ is $d \cdot m_2/k = 1$. Concretely, given a $\mathbb{Q}$-basis of $\mathcal{K}_2$ that expresses as $\{y, y^{\sigma_2}, \ldots, y^{\sigma_2^5}\}$, with $y \in \mathcal{K}_2$, then $\Lambda_{1,2}(\gamma)$ is computed as follows. The image under $\mathbf{A}$ decomposes as $\mathbf{A}(\gamma) \equiv \sum_{j=0}^5 \gamma_j y^{\sigma_2^j} \mod \ell\mathcal{O}$, and $\Lambda_{1,2}(\gamma) := \sum_{j=0}^5 \tilde{\gamma}_j\ \zeta^j \in \mathbb{Z}/\ell\mathbb{Z}$. Moreover, we have $\Lambda_{1,2}(\gamma^{\sigma_2}) = \zeta\Lambda_{1,2}(\gamma)$.
- **GSM on side $\mathcal{K}_1$.** The integer d only depends on k, hence $d = 1$ and the rank of $\Lambda_{1,1}$ is $d \cdot m_1/k = 2$. Given a $\mathbb{Q}$-basis of $\mathcal{K}_1$ that expresses as $\{y, y^{\sigma_1}, \ldots, y^{\sigma_1^5}, ay, ay^{\sigma_1}, \ldots, ay^{\sigma_1^5}\}$, with $y \in \mathcal{K}_1$ and $a \in \mathcal{K}_1^{\sigma_1}$, then $\Lambda_{1,1}(\gamma)$ is computed as follows. $\mathbf{A}$ is decomposed as $\mathbf{A}(\gamma) \equiv \sum_{j=0}^5 \gamma_{0,j} y^{\sigma_1^j} + \sum_{j=0}^5 \gamma_{1,j} ay^{\sigma_1^j} \mod \ell\mathcal{O}$. Then

$$\Lambda_{1,1}(\gamma) = \left(\sum_{j=0}^5 \gamma_{0,j}\ \zeta^j,\ \sum_{j=0}^5 \gamma_{1,j}\ \zeta^j\right) \in (\mathbb{Z}/\ell\mathbb{Z})^2.$$

  Moreover, we have $\Lambda_{1,1}(\gamma^{\sigma_1}) = \zeta\Lambda_{1,1}(\gamma)$.

5.2 Computing virtual logarithms with Galois Schirokauer maps

Consider Λ_d as in Definition 5 with d as announced in Corollary 2. Under Assumption 1, Λ_d is a Galois Schirokauer map as in Definition 4 with rank $\omega := d \cdot m/k$. If $\omega \geq r$, then the rank of Λ_d is large enough to represent all the units. Assume now $\omega < r$ and let us show a setup in which Λ_d is still convenient to compute a virtual logarithm map. Suppose the existence of a system of fundamental units $\{u_1, \ldots, u_r\}$ such that at least $r - \omega$ of them belong to $\cup_{s|k, s\neq k}\mathcal{K}^{\sigma^s}$. Rearranging the terms, assume that for all $\omega + 1 \leq i \leq r$,

$$\exists s|k,\ s \neq k, \quad \sigma^s(u_i) = u_i.$$

Then by Corollary 1 and Corollary 2, we have for all $\omega + 1 \leq i \leq r$,

$$\mathrm{vlog}(u_i) = 0 \quad \text{and} \quad \Lambda_d(u_i) = 0.$$

On the one hand, by the vanishing of the virtual logarithms, only the values of Λ_d on $\mathrm{span}(u_1, \ldots, u_\omega)$ are needed for computing the virtual logarithm map (see (3)). On the other hand, the vanishing of Λ_d fulfills exactly $\mathrm{rank}(\Lambda_d(\mathrm{span}(u_1, \ldots, u_r)) = \mathrm{rank}(\Lambda_d(\mathrm{span}(u_1, \ldots, u_\omega))$. In conclusion, in such a setup Λ_d defines a virtual logarithm map that fulfills Condition 1, which allows to accelerate the linear algebra step. The following proposition recapitulates the condition needed to accelerate the linear algebra step with the new GSM.

Proposition 4 (Conditions to accelerate the linear algebra step with GSM). *Let $(\mathcal{K}_1, \mathcal{K}_2)$ from Figure 3 of degree over $\mathbb{Q}$ equal to (m_1, m_2) with a TNFS automorphism (σ_1, σ_2) of order k. Consider $(\Lambda_{d,1}, \Lambda_{d,2})$ the corresponding GSM from Definition 5.*

Assume Assumption 1, hence the rank of $(\Lambda_{d,1}, \Lambda_{d,2})$ is equal to (ω_1, ω_2), where $\omega_i = d \cdot m_i/k$ for $i = 1, 2$ and $d = \prod_{p, \mathrm{val}_k(p)>0} p^{\mathrm{val}_p(k)-1}$. Suppose further that for $i = 1, 2$, there exists a system of fundamental units in $\mathcal{K}_i$ in which at most ω_i units do not belong to $\cup_{s|k, s\neq k}\mathcal{K}_i^{\sigma_i^s}$.

Then, $(\Lambda_{d,1}, \Lambda_{d,2})$ defines a virtual logarithm map that ensures Condition 1 and thus allows to speed up the linear algebra step by an approximate factor k^2.

5.3 Discussion about the speed-up factor

The linear system solved by the linear algebra step is represented in matrix form where each row corresponds to an equation, and each column corresponds to an unknown, either a virtual logarithm of an ideal (most columns), or a virtual logarithm of a fundamental units (very few columns). When the Conditions of Proposition 4 are met, we construct a virtual logarithm map satisfying for all prime ideals $\mathfrak{p}$ in the factor basis $\mathrm{vlog}(\mathfrak{p}^\sigma) = \zeta\,\mathrm{vlog}(\mathfrak{p})$, where ζ is a k-primitive root of unity and k is the automorphism's order. This allows to shrink the size of the linear system in the following way.

Let us first deal with "special ideals". If $\mathfrak{p}$ from the factor basis is fixed by σ^s for some s a proper divisor of k, then $\mathrm{vlog}(\mathfrak{p}^{\sigma^s}) = \zeta^s\,\mathrm{vlog}(\mathfrak{p}) = \mathrm{vlog}(\mathfrak{p})$, which implies $\mathrm{vlog}(\mathfrak{p}) = 0$ since $\zeta^s \neq 1$. The columns corresponding to such ideals are removed from the matrix, which shrinks its size already.

Now consider the most common situation where $\mathfrak{p}$ from the factor basis is not fixed by any σ^s for s a proper divisor of k. Denote $C_0, \ldots, C_{k-1}$ the columns corresponding to $\mathfrak{p}, \ldots, \mathfrak{p}^{\sigma^{k-1}}$. These k-columns are replaced by one column equal to $C := \sum_{i=0}^{k-1} \zeta^i C_i$, thus requiring a virtual logarithm map verifying $\mathrm{vlog}(\mathfrak{p}^\sigma) = \zeta\, \mathrm{vlog}(\mathfrak{p})$. In total, counting the "special ideals", the linear system size is reduced by factor larger than k.

To estimate the gain in performance in the linear algebra, it is crucial to examine the coefficient sizes of the new matrix and not only its dimension. Indeed, since the original matrix is sparse with coefficients mostly equal to -1, 0, or 1, Block-Wiedemann algorithm [10] accomplishes the linear algebra step in approximately λN^2 arithmetic operations, where λ designates the average number of non-zero coefficients per row and N the matrix dimension. The aforementioned operation reduces the matrix dimension to beneath N/k, while maintaining the average number of non-zero coefficients per row roughly at λ. Indeed, in most relations, a maximum of one ideal per orbit tends to appear. Further, the coefficients in the matrix become mostly equal to $\{-\zeta^i, 0, \zeta^i\}_{0 \leq i \leq k-1}$. The trick is to decompose the matrix $\mathcal{M}$ in basis ζ. Write $\mathcal{M} = \mathcal{M}_0 + \mathcal{M}_1\zeta + \cdots + \mathcal{M}_{k-1}\zeta^{k-1}$, where each matrix $\mathcal{M}_i$, with dimension approximately N/k, mostly contains coefficients equal to $\{-1, 0, 1\}$, with an average of non-zero coefficients per row around λ/k. Afterwards, each matrix-vector multiplication $\mathcal{M}v$ in the Wiedemann algorithm can be instead accomplished by k matrix-vector multiplications with the matrices $\mathcal{M}_i$, and $k \cdot N/k$ multiplications with k-roots of unity in $\mathbb{Z}/\ell\mathbb{Z}$:

$$\mathcal{M}v = \sum_{i=0}^{k-1} \zeta^i \left(\mathcal{M}_i v\right).$$

The number of arithmetic operations needed to perform a matrix-vector multiplication this way is $k \cdot \lambda/k \cdot (N/k) = \lambda \cdot N/k$ additions in $\mathbb{Z}/\ell\mathbb{Z}$ and $k \cdot N/k$ multiplications with k-roots of unity in $\mathbb{Z}/\ell\mathbb{Z}$. Counted in number of arithmetic operations, the cost of a matrix-vector multiplication comes down to approximately $(\lambda+k) \cdot N/k$. Overall, multiplying the cost of a matrix-vector multiplication by the matrix dimension we get the cost of the linear algebra using order k TNFS automorphisms, that is $(\lambda + k)(N/k)^2$. This needs to be compared with the cost of the linear algebra if the automorphisms are not used, which is λN^2. In conclusion, we expect an approximate performance improvement by a factor

$$\frac{\lambda}{\lambda + k} \cdot k^2. \tag{4}$$

Since the order k ($k = 6, 12$ in our applications) is small compared to λ ($\lambda = 200$ in a record-size computation), we approximate the acceleration factor to k^2.

6 Application: Acceleration of the linear algebra step with order 6 and 12 TNFS automorphisms

Let $\mathbb{F}_{p^6}$ (resp. $\mathbb{F}_{p^{12}}$) any finite field of characteristic p and extension degree 6 (resp. 12). We show in this section how to construct a TNFS diagram on $\mathbb{F}_{p^6}$ (resp. $\mathbb{F}_{p^{12}}$) using the *Conjugation* polynomial selection method and ensuring two properties. On the one hand, the diagram enjoys an order 6 (resp. 12) TNFS automorphism, and on the other hand, the new GSM from Definition 5 fulfills the conditions of Proposition 4. As a consequence, the TNFS automorphism can be used to accelerate the relation collection by an approximate factor of 6 (resp. 12) and the linear algebra step by an approximate factor of 36 (resp. 144).

6.1 Construction of order 6 and 12 TNFS automorphisms

The extension degree n is 6 or 12. It decomposes as $n = \eta\kappa$ with η and κ coprime and non-trivial (hence two possible setups for $n = 6$, and two setups for $n = 12$). In all these setups, the Conjugation method presented in §2.1 with the choices of Table 4 provides a TNFS diagram with two automorphisms $\sigma_1 \in \mathrm{Aut}(\mathcal{K}_1)$ and $\sigma_2 \in \mathrm{Aut}(\mathcal{K}_2)$, both of order n. The following proposition states that they constitute a TNFS automorphism as in Definition 3.

Proposition 5. *For each of the setups $(\eta = 3, \kappa = 2)$ or $(\eta = 2, \kappa = 3)$ for $n = 6$, or $(\eta = 3, \kappa = 4)$ or $(\eta = 4, \kappa = 3)$ for $n = 12$, let h, f_1 and f_2 three polynomials selected by the Conjugation method, where h of degree η has cyclic Galois group and, f_1 and f_2 of respective degrees 2κ and κ are selected with Table 4. Additionally, suppose that p does not divide their discriminant. Denote $\sigma_1 \in \mathrm{Aut}(\mathcal{K}_1)$ and $\sigma_2 \in \mathrm{Aut}(\mathcal{K}_2)$ the resulting order n automorphisms. Then they each fix a prime ideal of degree n above p, that is, (σ_1, σ_2) is a TNFS automorphism.*

Proof. Denote σ_h, σ_{f_1} and σ_{f_2} of respective orders η, κ, and κ, the automorphisms corresponding to the polynomials. Recall that since η and κ are coprime, the automorphisms σ_1 and σ_2 are defined by the joint action of σ_h and σ_{f_1}, and σ_h and σ_{f_2}. It is sufficient to prove that σ_h fixes a prime ideal of degree η above p, and σ_{f_1} and σ_{f_2} each fixes a prime ideal of degree κ above p.

Denote $\mathcal{O}_h$, $\mathcal{O}_{f_1}$, and $\mathcal{O}_{f_2}$ the ring of integers of $\mathcal{K}_h$, $\mathcal{K}_{f_1}$ and $\mathcal{K}_{f_2}$. By construction, h is irreducible modulo p, which means that $p\mathcal{O}_h$ is irreducible of degree η. Therefore, $p\mathcal{O}_h$ is fixed by σ_h since the conjugate of a prime ideal above p is a prime ideal above p. The same scenario occurs in $\mathcal{K}_{f_2}$, the automorphism σ_{f_2} fixes $p\mathcal{O}_{f_2}$ which is an irreducible prime ideal of degree κ.

It remains to prove that σ_{f_1} fixes a prime ideal of degree κ above p. We know that $p\mathcal{O}_{f_1}$ is not prime. In fact, by construction f_1 has an irreducible factor of degree κ modulo p, hence, $p\mathcal{O}_{f_1}$ splits into a degree κ prime ideal $\mathfrak{p}$ times a remaining part of total degree κ. If the remaining part is not a prime ideal of degree κ, then $\sigma(\mathfrak{p}) = \mathfrak{p}$ since the conjugate of a prime ideal above p is a prime ideal above p with the *same* residual degree. Hence, we restrict to the case $p\mathcal{O}_{f_1} = \mathfrak{p}\mathfrak{q}$, with $\mathfrak{p}$ and $\mathfrak{q}$ are both irreducible prime ideals of degree κ. A root of the polynomial μ (from the Conjugation method in §2.1) in $\mathcal{K}_{f_1} := \mathbb{Q}(\alpha_1)$ is $-g_0(\alpha_1)/g_1(\alpha_1)$ which is stable under the action of σ_1 for all the choices of (g_0, g_1) in Table 4. Therefore, the number field $\mathcal{K}_\mu := \mathbb{Q}[x]/(\mu)$ is isomorphic to $\mathcal{K}_{f_1}^{\sigma_{f_1}}$, and σ_{f_1} is a $\mathcal{K}_\mu$-automorphism of $\mathcal{K}_{f_1}$. We conclude the proof by looking at the decomposition pattern of the characteristic p along the tower $\mathbb{Q} \subset \mathcal{K}_\mu \subset \mathcal{K}_{f_1}$. Denote $\mathcal{O}_\mu$ the ring of integer of $\mathcal{K}_\mu$. By construction, μ has two roots modulo p, which means that $p\mathcal{O}_\mu$ decomposes as a product of two prime ideals $\mathfrak{p}_a$ and $\mathfrak{p}_b$, each of residual degree 1. Further, recall that $p\mathcal{O}_{f_1} = \mathfrak{p}\mathfrak{q}$, therefore, after reordering the ideals, we have $\mathfrak{p}_a\mathcal{O}_{f_1} = \mathfrak{p}$ and $\mathfrak{p}_b\mathcal{O}_{f_1} = \mathfrak{q}$. Since σ_{f_1} is a $\mathcal{K}_\mu$-automorphism of $\mathcal{K}_{f_1}$, we must have $\sigma_{f_1}(\mathfrak{p})$ a prime ideal above $\mathfrak{p}_a$, thus equal to $\mathfrak{p}$. It is noteworthy that the reliance on the choice of (g_0, g_1) is not required if κ is odd. Indeed, we have $\sigma_{f_1}(\mathfrak{p}) = \mathfrak{p}$ as the alternative $\sigma_{f_1}(\mathfrak{p}) = \mathfrak{q}$ would imply that $\sigma_{f_1}^{\kappa}(\mathfrak{p}) = \mathfrak{q}$. This contradicts the fact that the order of σ_{f_1} is κ.

6.2 Large Enough Rank for the Galois Schirokauer Map

Once the TNFS diagram is constructed with an order n TNFS automorphism (σ_1, σ_2) as described in Proposition 5 where $n = 6$ or $n = 12$, Construction 5 defines a GSM equal to

- $(\Lambda_{1,1}, \Lambda_{1,2})$ of rank $(2, 1)$ if $n = 6$,
- $(\Lambda_{2,1}, \Lambda_{2,2})$ of rank $(4, 2)$ if $n = 12$.

To satisfy the conditions of Proposition 4, we need the number of fundamental units not belonging to subfields related to the automorphism to be bounded by the rank of the Schirokauer map. In Theorem 1, we provide restrictions on the number of real roots of the selected polynomials to ensure that such conditions are always met for $n = 6$ and $n = 12$. The following lemma provides a formula to compute the unit rank of a compositum.

Lemma 3 (Unit rank of compositum of number fields). *Let $\mathcal{K}$ and $\mathcal{L}$ two lineary disjoint number fields over Q with degrees n and m and signatures (u, v) and (μ, ν). Then the unit rank of their compositum $\mathcal{M}$ is $1/2(nm + u\mu) - 1$.*

Proof. The embeddings of $\mathcal{M}$ are $\{(g, l) | g$ embedding of $\mathcal{K}_h, l$ embedding of $\mathcal{K}_{f_i}\}$ because $\mathcal{K}$ and $\mathcal{L}$ are lineary disjoint. Therefore, the number of real embeddings of $\mathcal{M}$ is $u \times \mu$, and the lemma follows from Dirichlet's unit theorem.

In TNFS context, for $i = 1, 2$, the middle number field $\mathcal{K}_i$ is the compositum of $\mathcal{K}_h$ and $\mathcal{K}_{f_i}$, which are lineary disjoint over $\mathbb{Q}$ since $\deg_{\mathbb{Q}}(\mathcal{K}_i) = \deg_{\mathbb{Q}}(\mathcal{K}_h) \times \deg_{\mathbb{Q}}(\mathcal{K}_{f_i})$. By Lemma 3 we have

$$r_{\mathcal{K}_i} = 1/2\,(\eta \deg(f_i) + r_h r_{f_i}) - 1, \tag{5}$$

where r_f denotes the number of real roots of a polynomial f. We also need the following assumption which is related to Leopoldt's conjucture [3] (in the sens that Leopoldt's conjucture is the p-adic version of the following assumption).

Assumption 2 (Unit lift assumption) *Let $\mathcal{K} \subset \mathcal{L}$ an extension of number fields and $\{u_1, \dots, u_r\}$ a system of fundamental units of $\mathcal{K}$. Denote $\tilde{u}_i$ the lift of u_i to $\mathcal{L}$ for $1 \leq i \leq r$. Then $\{\tilde{u}_1, \dots, \tilde{u}_r\}$ extends to a system of fundamental units of $\mathcal{L}$.*

Theorem 1 (Speed-up of the linear algebra step for $n = 6$ and $n = 12$). *Let $n = 6$ or $n = 12$. Let h, f_1 and f_2 constructed with the Conjugation polynomial selection with an order n TNFS automorphism (σ_1, σ_2) as described in Proposition 5, hence p does not divide their discriminant. Suppose further the following restrictions on the number of real roots of h, f_1 and f_2:*

1. ***If** η **is even**, then h has no real roots.*
2. ***If** η **is odd**, then f_1 and f_2 have no real roots.*

Table 5 recapitulates these restrictions. Then, under Assumption 2, the field $\mathcal{K}_1$ (resp. $\mathcal{K}_2$) admits a system of fundamental units with:

- *At most 2 (resp. 1) fundamental units that do not belong to $\mathcal{K}_1^{\sigma_1^2} \cup \mathcal{K}_1^{\sigma_1^3}$ (resp. $\mathcal{K}_2^{\sigma_2^2} \cup \mathcal{K}_2^{\sigma_2^3}$) if $n = 6$.*
- *At most 4 (resp. 2) fundamental units that do not belong to $\mathcal{K}_1^{\sigma_1^6} \cup \mathcal{K}_1^{\sigma_1^4}$ (resp. $\mathcal{K}_2^{\sigma_2^6} \cup \mathcal{K}_2^{\sigma_2^4}$) if $n = 12$.*

Consequently, the GSM of Definition 5 satisfy the conditions of Proposition 4, i.e., it allows to speed-up the linear algebra step by a factor approximately equal to 36 if $n = 6$, and 144 if $n = 12$.

n	6		12	
Setup	$(\eta = 2, \kappa = 3)$	$(\eta = 3, \kappa = 2)$	$(\eta = 3, \kappa = 4)$	$(\eta = 4, \kappa = 3)$
Requirement: Polynomials with no real roots	h	f_1 and f_2	f_1 and f_2	h

Table 5: Requirements on the polynomials output by the Conjugation method.

Proof. Assumption 2 reduces the conditions on the distribution of fundamental units along subfields to requirements on unit ranks. Precisely, we shall prove

- **If** $n = 6$**,** then $r_{\mathcal{K}_i} - (r_{\mathcal{K}_i^{\sigma_i^3}} + r_{\mathcal{K}_i^{\sigma_i^2}} - r_{\mathcal{K}_i^{\sigma_i}}) \leq w_i$, where $(w_1, w_2) = (2, 1)$.
- **If** $n = 12$**,** then $r_{\mathcal{K}_i} - (r_{\mathcal{K}_i^{\sigma_i^6}} + r_{\mathcal{K}_i^{\sigma_i^4}} - r_{\mathcal{K}_i^{\sigma_i^2}}) \leq w_i$ where $(w_1, w_2) = (4, 2)$.

The theorem depends on the subfields trellis of the number fields $\mathcal{K}_1$ and $\mathcal{K}_2$. For this reason we consider each setup. In particular, we prove the theorem separately when $n = 6$ and when $n = 12$.

The extension degree $n = 6$. Recall that the ranks of the Schirokauer maps on $(\mathcal{K}_1, \mathcal{K}_2)$ are $(1, 2)$. Let us consider the two setups $\eta = 3$ and $\eta = 2$.

1. **The case $\eta = 3$ represented in Figure 6.** Suppose η is taken equal to 3. The polynomial h is chosen such that $\mathcal{K}_h$ is Galois. Therefore, $\mathcal{K}_h$ is totally real (since its degree is odd), and its unit rank is equal to 2. The rest of the proof is divided into the examination of the units in $\mathcal{K}_1$ and in $\mathcal{K}_2$.
   - **Distribution of the units along the subfields of $\mathcal{K}_2$.** Since h has three real roots, the unit rank of $\mathcal{K}_2$ is $r_{\mathcal{K}_2} = 2 + 3r_{f_2}/2$. The field $\mathcal{K}_2$ has two subfields related to σ_2 that are $\mathcal{K}_2^{\sigma_2^2} \simeq \mathcal{K}_{f_2}$ and $\mathcal{K}_2^{\sigma_2^3} \simeq \mathcal{K}_h$ with respective absolute degrees 2 and 3. By Dirichlet's theorem, the sum of their unit ranks is $2 + r_{f_2}/2$. Further, the subfield $\mathcal{K}_2^{\sigma_2^2} \cap \mathcal{K}_2^{\sigma_2^3} = \mathcal{K}^{\sigma}$ is isomorphic to $\mathbb{Q}$, hence its unit rank is 0. The required condition becomes $2 + 3r_{f_2}/2 - (2 + r_{f_2}/2) \leq 1$, which is equivalent to $r_{f_2} \leq 1$. Since f_2 has degree 2 and the number of real roots of f_2 is necessarily even, this is equivalent to requiring f_2 with no real roots.
   - **Distribution of the units along the subfields of $\mathcal{K}_1$.** The field $\mathcal{K}_1$ has unit rank $r_{\mathcal{K}_1} = 5 + 3r_{f_1}/2$. Further, its subfield $\mathcal{K}_1^{\sigma_1^2} \simeq \mathcal{K}_{f_1}$ has absolute degree 4 and unit rank $r_{f_2}/2 + 1$. The subfield $\mathcal{K}_1^{\sigma_1^3}$ has two lineary independent subfields, $\mathcal{K}_h$ and $\mathcal{K}_1^{\sigma_1}$ (since there absolute degrees that are 3 and 2 are coprime). That is to say $r_{\mathcal{K}_1^{\sigma_1^3}} \geq r_{\mathcal{K}_1^{\sigma_1}} + r_{\mathcal{K}_h}$, where $r_{\mathcal{K}_h} = 2$. Therefore, $r_{\mathcal{K}_1} - (r_{\mathcal{K}_1^{\sigma_1^3}} + r_{\mathcal{K}_1^{\sigma_1^2}} - r_{\mathcal{K}_1^{\sigma_1}}) \leq 2 + 2r_{f_1}$. Hence requiring $r_{f_1} = 0$ fulfills the announced condition.
2. **The case $\eta = 2$ represented in Figure 7** Here η is set to 2. The unit rank of $\mathcal{K}_h$ is equal to 0 since h has no real roots. The rest of the proof is divided into the examination of the unit distribution in $\mathcal{K}_1$ and in $\mathcal{K}_2$.
   - **Distribution of the units along the subfields of $\mathcal{K}_2$.** The polynomial f_2 has 3 real roots since it has odd degree 3 and an automorphism of order 3. Thus, by (5), the unit rank of $\mathcal{K}_2$ is $3r_h/2 + 2 = 2$. The sum of the unit ranks of $\mathcal{K}_2^{\sigma_2^2} \simeq \mathcal{K}_h$ and $\mathcal{K}_2^{\sigma_2^3} \simeq \mathcal{K}_{f_2}$ is $r_h/2 + 2 = 2$. Since $\mathcal{K}_2^{\sigma_2} \simeq \mathbb{Q}$, we get $r_{\mathcal{K}_2} - (r_{\mathcal{K}_2^{\sigma_2^3}} + r_{\mathcal{K}_2^{\sigma_2^2}} - r_{\mathcal{K}_2^{\sigma_2}}) = 0 \leq 1$.
   - **Distribution of the units along the subfields of $\mathcal{K}_1$.** Since h has no real roots, then by Equation (5), the unit rank of $\mathcal{K}_1$ is 5. Let us distinguish two cases whether the extension field $\mathcal{K}_1^{\sigma_1}$, which is of absolute degree 2, is totally real or totally imaginary. Suppose first that $\mathcal{K}_1^{\sigma_1}$ is totally real. The field $\mathcal{K}_1^{\sigma_1^3} \simeq \mathcal{K}_{f_1}$ is a relative extension 3 field over $\mathcal{K}_1^{\sigma_1}$. Thus, $\mathcal{K}_1^{\sigma_1^3}$ cannot be totally complex. Its units rank can only take values equal to 3, 4, or 5 (not 2). In all three cases, we have $r_{\mathcal{K}_1} - (r_{\mathcal{K}_1^{\sigma_1^3}} + r_{\mathcal{K}_1^{\sigma_1^2}} - r_{\mathcal{K}_1^{\sigma_1}}) \leq r_{\mathcal{K}_1} - r_{\mathcal{K}_1^{\sigma_1^3}} \leq 5 - 3 = 2$. Suppose now that $\mathcal{K}_1^{\sigma_1}$ is totally imaginary. Then, the unit rank of $\mathcal{K}_1^{\sigma_1}$ is 0. The fields $\mathcal{K}_1^{\sigma_1^3}$ and $\mathcal{K}_1^{\sigma_1^2}$ have respectively unit ranks at least 2 and 1 since their absolute degrees are

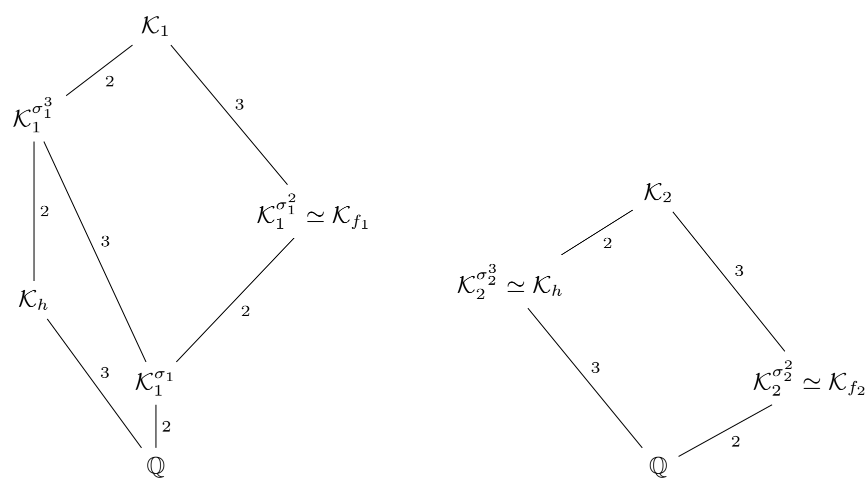
6 and 4. Hence we count at least 3 independent units in $\mathcal{K}_1^{\sigma_1^2} \cup \mathcal{K}_1^{\sigma_1^3}$ out of the 5 units of $\mathcal{K}_1$.

The extension degree $n = 12$. Recall that the ranks of the Schirokauer maps on $(\mathcal{K}_1, \mathcal{K}_2)$ are $(2, 4)$. Consider the two setups $\eta = 3$ and $\eta = 4$.

- **The case $\eta = 3$ represented in Figure 8.** The polynomial h has three real roots and the unit rank of $\mathcal{K}_h$ is 2. Let us examine the distribution of the units along the subfields of $\mathcal{K}_2$ and $\mathcal{K}_1$.
  - **Distribution of the units along the subfields of $\mathcal{K}_2$.** The unit rank of $\mathcal{K}_2$ is $5 + 3r_{f_1}/2$. We require f_2 to have no real roots, hence, $r_{\mathcal{K}_2} = 5$. Further, we count 2 for the unit rank of $\mathcal{K}_h$ which is isomorphic to a subfield of $\mathcal{K}_2^{\sigma_2^6}$, and 1 for the unit rank of $\mathcal{K}_{f_2} \simeq \mathcal{K}_2^{\sigma_2^4}$. Since the absolute degrees of $\mathcal{K}_h$ and $\mathcal{K}_2^{\sigma_2^2}$ (that are 3 and 2) are coprime, we necessarily have $r_{\mathcal{K}_2^{\sigma_2^6}} + r_{\mathcal{K}_2^{\sigma_2^4}} - r_{\mathcal{K}_2^{\sigma_2^2}} \geq r_{\mathcal{K}_h} + r_{\mathcal{K}_2^{\sigma_2^4}} \geq 3$. The required equation follows.
  - **Distribution of the units along the subfields of $\mathcal{K}_1$.** We require f_1 to have no real roots. Hence, the unit rank of $\mathcal{K}_1$ is 11. On the one hand, the unit rank of $\mathcal{K}_1^{\sigma_1^4} \simeq \mathcal{K}_{f_1}$ is 3. On the other hand, we shall count the unit ranks of $\mathcal{K}_1^{\sigma_1^6}$ and the intersection $\mathcal{K}_1^{\sigma_1^4} \cap \mathcal{K}_1^{\sigma_1^6} = \mathcal{K}_1^{\sigma_1^2}$. Let $(r_g, .)$ the signature of $\mathcal{K}_1^{\sigma_1^2}$. Since its absolute degree is 4, its unit rank is equal to $1 + r_g/2$. Further, $\mathcal{K}_1^{\sigma_1^6}$ is the compositum of $\mathcal{K}_1^{\sigma_1^2}$ and $\mathcal{K}_h$ which have coprime absolute degrees 4 and 3. Consequently by (5), the unit rank of $\mathcal{K}_1^{\sigma_1^6}$ is $5 + 3r_g/2$. Therefore, we have $r_{\mathcal{K}_1^{\sigma_1^6}} + r_{\mathcal{K}_1^{\sigma_1^4}} - r_{\mathcal{K}_1^{\sigma_1^2}} = 7 + r_g$. Since $r_g \geq 0$, and the unit rank of $\mathcal{K}_1$ is 11, we have the required $r_{\mathcal{K}_1} - (r_{\mathcal{K}_1^{\sigma_1^6}} + r_{\mathcal{K}_1^{\sigma_1^4}} - r_{\mathcal{K}_1^{\sigma_1^2}}) \leq 4$.
- **The case $\eta = 4$ represented in Figure 9.** We require the polynomial h to have no real roots. Therefore, the unit rank of $\mathcal{K}_h$ is 1. We continue the proof by examining the units in both number fields $\mathcal{K}_2$ and $\mathcal{K}_1$.
  - **Distribution of the units along the subfields of $\mathcal{K}_2$.** The unit rank of $\mathcal{K}_2$ is 5. Since f_2 has an automorphism of order 3, the unit rank of $\mathcal{K}_{f_2} \simeq \mathcal{K}_2^{\sigma_2^3}$ is 2. Additionally, we count 1 for the unit rank of $\mathcal{K}_h \simeq \mathcal{K}_2^{\sigma_2^4}$. Since these two subfields have coprime absolute degrees equal respectively to 3 and 4, the required equation follows.
  - **Distribution of the units along the subfields of $\mathcal{K}_1$.** Since h has no real roots, the unit rank of $\mathcal{K}_1$ is 11. We want to prove that the quantity $\nu := r_{\mathcal{K}_1^{\sigma_1^6}} + r_{\mathcal{K}_1^{\sigma_1^4}} - r_{\mathcal{K}_1^{\sigma_1^2}}$ is greater or equal to $11 - 4 = 7$. The absolute degree of $\mathcal{K}_1^{\sigma_1^4}$ is 8, hence unit rank is at least 3. Further, let $(e, .)$ the signature of $\mathcal{K}_1^{\sigma_1^2}$, then its unit rank is $1 + e/2$. Moreover, consider a polynomial U that defines $\mathcal{K}_1^{\sigma_1^6}$ over $\mathcal{K}_1^{\sigma_1^2}$, i.e, $\mathcal{K}_1^{\sigma_1^6} = \mathcal{K}_1^{\sigma_1^2}[X]/(U)$. The polynomial U can be taken with rational coefficients since its degree 3 is coprime with the absolute degree of $\mathcal{K}_1^{\sigma_1^2}$ that is 4. Since the number of

real roots of U is at least 1, Equation (5) implies the unit rank of $\mathcal{K}_1^{\sigma_1^6}$ is at least $5 + e/2$. Overall, we get that $\nu \geq 5 + e/2 + 3 - (1 + e/2) = 7$.

In conclusion, it is sufficient to restrict the number of real roots of the selected polynomials as indicated in Table 5, which is practically easy, in order to ensure that the GSM construction from Definition 5 is sufficient to compute discrete logarithms while using order 6 or 12 TNFS automorphisms to accelerate the linear algebra step. These restrictions allow to construct the number fields $\mathcal{K}_1$ and $\mathcal{K}_2$ with enough fundamental units that belong to subfields related to the automorphisms, and therefore, the few remaining units that do not vanish by both the GSM and the virtual logarithm can be controlled by the GSM.



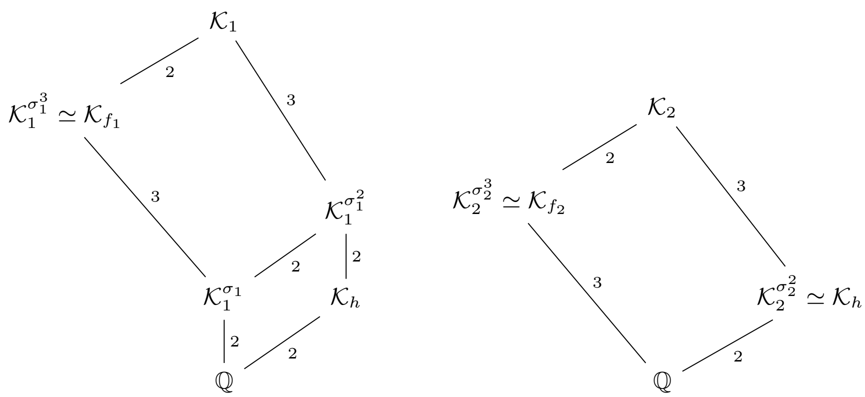
(a) Lattice of σ_1-related subfields of $\mathcal{K}_1$ of absolute degree 12.

(b) Lattice of σ_2-related subfields of $\mathcal{K}_2$ of absolute degree 6.

Fig. 6: Lattices of automorphism-related subfields with the setup $n = 6$ and $\eta = 3$. Each edge indicates a field extension and is labeled with the corresponding extension degree.

6.3 Application to other Automorphism Orders?

The natural continuation of this work is to extend the use of Galois Schirokauer maps to other composite orders than 6 and 12. The dimensions of the GSM are provided by Corollary 2 and their values are exhibited in Table 10 for various values of n. If some restrictions on the polynomials restrict enough fundamental units to belong to subfields related to the automorphism, then theses GSM allow



(a) Lattice of σ_1-related subfields of $\mathcal{K}_1$ of absolute degree 12.

(b) Lattice of σ_2-related subfields of $\mathcal{K}_2$ of absolute degree 6.

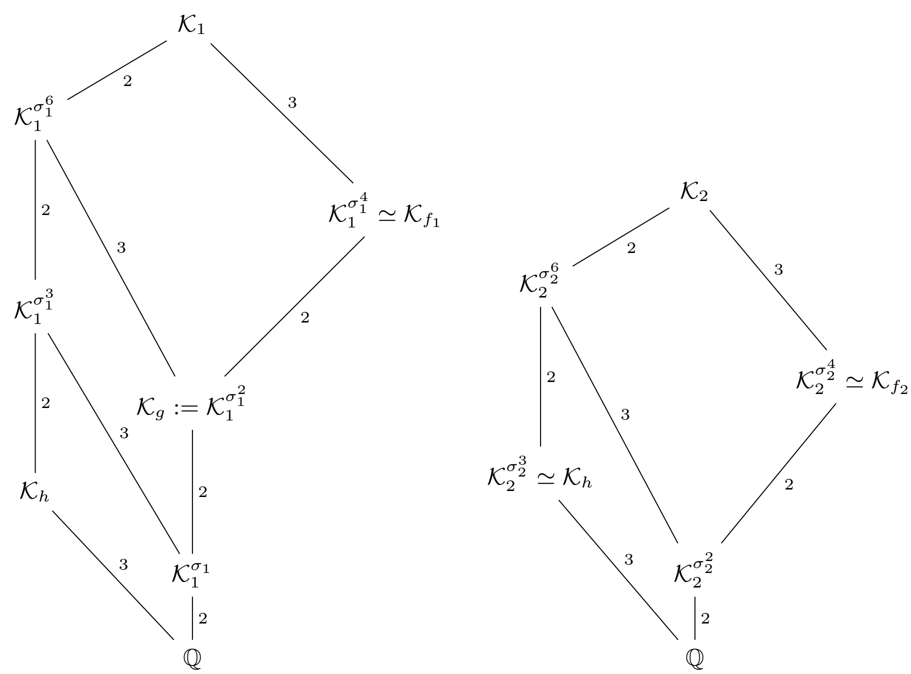
Fig. 7: Lattices of automorphism-related subfields with the setup $n = 6$ and $\eta = 2$. Each edge indicates a field extension and is labeled with the corresponding extension degree.

to accelerate the linear algebra step with approximately a factor n^2. The problem of constructing a TNFS automorphism of degree n for any extension degree n is an open problem. We were not able to solve it for integers n that are powers of primes (e.g. 4, 9, 16), where κ and η cannot be taken coprime. For other degrees that are square-free such as 10, 14, 15, while the construction of TNFS automorphisms might be feasible, our GSM rank is too low. The investigation of other non square-free orders such as 18, 20 and the powers of prime ones is left for a future work. Another interesting continuation is to apply our GSM to other polynomial selection methods.

n	4	6	8	9	10	12	14	15
Ranks	$(4, 2)$	$(2, 1)$	$(8, 4)$	$(6, 3)$	$(2, 1)$	$(4, 2)$	$(2, 1)$	$(2, 1)$

n	16	18	20	21	22	24	25	26
Ranks	$(16, 8)$	$(6, 3)$	$(4, 2)$	$(2, 1)$	$(2, 1)$	$(8, 4)$	$(10, 5)$	$(2, 1)$

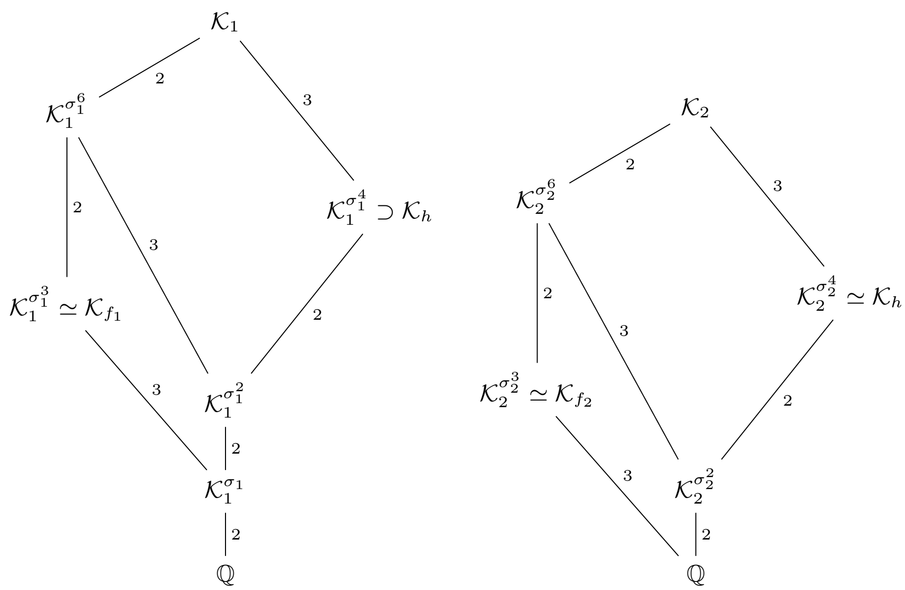
Table 10: Ranks of the GSM of Definition 5 on $(\mathcal{K}_1, \mathcal{K}_2)$ of absolute degrees $(2n, n)$, and constructed with an order n TNFS automorphism.



(a) Lattice of σ_1-related subfields of $\mathcal{K}_1$ (absolute degree 24).

(b) Lattice of σ_2-related subfields of $\mathcal{K}_2$ (absolute degree 12).

Fig. 8: Lattices of automorphism-related subfields with the setup $n = 12$ and $\eta = 3$. Each edge indicates a field extension and is labeled with the corresponding extension degree.



(a) Lattice of σ_1-related subfields of $\mathcal{K}_1$ (absolute degree 24).

(b) Lattice of σ_2-related subfields of $\mathcal{K}_2$ (absolute degree 12).

Fig. 9: Lattices of automorphism-related subfields with the setup $n = 12$ and $\eta = 4$. Each edge indicates a field extension and is labeled with the corresponding extension degree.

7 Experimental validation: acceleration of the linear algebra step

To validate our findings, we provide a Sagemath implementation of the Tower Number Field Sieve in which the user can choose to use the usual Schirokauer map, or our construction of Galois Schirokauer map together with TNFS automorphisms. We demonstrate our results by applying TNFS on degree 6 and 12 finite fields. Our implementation is available at [3] together with the complete data of our experiments (polynomials, TNFS parameters ...). We emphasize that our implementation is not optimized, in fact, all presented examples in this section can be solved with elementary algorithms. The goal of our implementation is twofold. First, it shows that our theoretical findings work and allow to compute discrete logarithms. Second, it demonstrates the factor 6 and 12 reduction of the size of the matrix in TNFS brought by our work. These results are presented in Table 11. Using automorphisms or not, no example worked with the setup ($n = 12$, $\eta = 3$, $\kappa = 4$). Our explanation is that the sieve dimension that is $2\eta = 6$ is too low given that the characteristic size is small. To make such examples work, it is necessary to consider sieve elements with degree larger than 1 over $\mathcal{K}_h$, which is not supported by our code, or to increase the characteristic size, thus reaching values that are out of the scope of what Sagemath can handle. The implementation works in the following steps.

Finite field	bitsize of p^n	Setup: $\eta =$	(Matrix dimension), rank	
			Without automorphism	**Our work**
$\mathbb{F}_{29^6}$	30	2	$(34686 \times 24006), 23334$	$(5849 \times 4001),\ 3878$
$\mathbb{F}_{37^6}$	32	3	$(1177 \times 510), 502$	$(202 \times 85),\ 81$
$\mathbb{F}_{53^6}$	35	2	*	$(15580 \times 9874),\ 9666$
$\mathbb{F}_{17^{12}}$	50	4	$(86934 \times 23196), 22647$	$(7311 \times 1936),\ 1874$
$\mathbb{F}_{53^{12}}$	69	4	*	$(22674 \times 15469),\ 15291$

Table 11: Experiment results. Comparison of the matrix size between the use or not of TNFS automorphisms of orders 6 and 12. A * indicates that SageMath killed the computation because of the large matrix size. The complete data of these computations are provided in the git repository.

Polynomial selection. For $n = 6, 12$, to perform a computation on a finite field $\mathbb{F}_{p^n}$ of characteristic p, the polynomials must be selected following the Conjugation method presented §2.1 and Table 4. This ensures the presence of an order n TNFS automorphism as proved in Propositions 5 Moreover, the polynomials must be chosen with the constraints on the number of real roots of Theorem 1 which are recapitulated in Table 5. The choice of the polynomials

[3] `gitlab.inria.fr/halaswad/accelerating_tnfs_with_galois_automorphisms`

is done by the method `polyselect` in `tnfs.py`. The user must set adequate polynomials $(h, f_1, \mu \dots)$ in the method `set_from_family`.

Relation collection. We use the special-q technique to collect relations [31]. Only one special-q ideal is considered per orbit. Thereafter, if the automorphisms are not used in the code, we simulate their "non-use" by expanding each relation into n conjugate relations which is done with the method `expand_list_phi`. If the automorphisms are used, then we do not expand the relations and proceed to the next step. This is carried by the method `relation_collection` in `tnfs.py`.

Linear algebra. We construct the GSM $(\Lambda_{d_n,1}, \Lambda_{d_n,2})$ from Definition 5, where $d_6 = 1$ and $d_{12} = 2$. Then, via matrix multiplication in our code, we replace each n conjugate columns by one column, and remove the columns corresponding to prime ideals fixed by a power of the automorphism, as explained in Section 5.3. Consequently, the number of columns is reduced by a factor slightly larger than n and the number of rows by a factor n (we need to find n less relations). We did not implement the Wiedemann algorithm in Sagemath. We rather use Sagemath functions to solve the linear system and check that our algorithm computes a virtual logarithm map as expected. Therefore, we only compare the sizes of the matrices depending on whether the order n TNFS automorphism is deployed or not. Table 11 presents the matrices sizes on several finite fields of degree 6 and 12. In the `tnfs.py` file, the methods `relation_matrices` constructs the matrix of relations, and the methods `linear_algebra` and `modified_linear_algebra` find a kernel element of the matrix. The first applies when the automorphisms are not used, and the second when they are used. Integrating the GSM constructions in the cado-nfs software [1] and modifying the linear algebra to deal with the matrix decomposition in basis ζ explained in Section 5.3 is left for a future work.

Final step : verification. While we did not implement the *individual logarithm* step—as our work does not improve it—we ensured the consistency of the virtual logarithm map computed by our linear algebra step. This was achieved by considering numerous sieve elements ϕ in one of the two middle number fields that correspond to relations from previous steps. Subsequently, we confirmed that their virtual logarithms correspond to the logarithms, in some basis $\tilde{g}$, of their projection to the finite field. This is done as follows.

We choose g a generator of the sub-group of order ℓ of the finite field's multiplicative group. For each considered ϕ, we compute on the one hand its virtual logarithm $\mathrm{vlog}(\phi)$ using the output of our algorithm, and on the other hand its logarithm $\log_g(\phi)$ in basis g using the logarithm function implemented in Sagemath. While the quotient $\mathrm{vlog}(\phi)/\log_g(\phi) \bmod \ell$ is constant over the picked ϕ's, say equal to c, we continue with different sieve elements. If a sieve element provides a quotient different from c, then the algorithm failed. If the quotient is constant over all the $\phi's$ (in number of 40 in each of our experiments), then we assume that the algorithm succeeded. The function `log_consistency_check` within `tnfs.py` performs this check.

7.1 Performing a computation

In the following B, q_0, and q_1 denote respectively the smoothness bound, the smallest special-q prime and the largest special-q prime used. In each special-q lattice, the search for relations was performed in a ball of dimension $2 \times \eta$ and with a radius `rad`. The radius is set according to the input E so that the total search space is approximately of the size of the hypercube $[-E, E]^{2\cdot\eta}$. See the document `tnfs.py` for the exact expression of `rad`. Moreover, in some of the computations a *line-sieve* is performed to factor all elements ϕ with coefficients sizes bounded by the new parameter El. This allows to find relations that do not exhibit prime ideals belonging to the special-q range.

The reader can perform a computation by running the `play_tnfs.sage`. The variable `attempt` sets the parameters of the example that will be run, the boolean `use_auto` indicates to use or ignore the TNFS automorphism and the integer d must be set according to Corollary 2. Hence, the variable d must be set to 1 when employing an order 6 automorphism and to 2 when the order of the automorphism is 12.

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A Proofs of Proposition 1 and Proposition 3

A.1 Proof of Proposition 1

First, each of the three sets Γ/Γ^ℓ, $\mathcal{O}^\times/(\mathcal{O}^\times)^\ell$, and $\mathcal{I}/\mathcal{I}^\ell$ is clearly an $\mathbb{F}_\ell$-vector space where the underlying scalar product is $(\lambda, x) \mapsto x^\lambda$. To prove the isomorphism statement, it is sufficient to prove that we have the following exact sequence and that it splits:

$$1 \longrightarrow \mathcal{O}^\times/(\mathcal{O}^\times)^\ell \xrightarrow{i} \Gamma/\Gamma^\ell \xrightarrow{j} \mathcal{I}/\mathcal{I}^\ell \longrightarrow 1.$$

In the following, an element x is written $\overline{x}$ when considered modulo ℓ-th powers.

The morphism i is injective. The morphism i is derived from $\mathcal{O}^\times \subset \Gamma$. To prove that it is one-to-one, consider $x \in \mathcal{O}^\times$ such that $i(\overline{x}) = \overline{1}$. That is, there exists $\gamma \in \Gamma$ such that $x\gamma^\ell = 1$. Then, γ^ℓ belongs to $\mathcal{O}$, and so do γ. Indeed, if $P(X)$ is a monic polynomial with integer coefficients vanishing on γ^ℓ, then $P(X^\ell)$ is a monic polynomial with integer coefficients vanishing on γ. Furthermore, γ belongs to $\mathcal{O}^\times$ since $\gamma(x\gamma^{\ell-1}) = 1$. In conclusion we have $x \equiv 1 \bmod (\mathcal{O}^\times)^\ell$.

The morphism j is surjective. The morphism j is defined by $\gamma \bmod \Gamma^\ell \mapsto (\gamma) \bmod \mathcal{I}^\ell$. It is clearly well defined, let us prove its surjectivity. Since the class number $h_\mathcal{K}$ is coprime to ℓ, let $a, b \in \mathbb{Z}$ such that $ah_\mathcal{K} + b\ell = 1$. Consider $I \in \mathcal{I}$. We have

$$I = I^{ah_\mathcal{K}+b\ell} = (I^a)^{h_\mathcal{K}}(I^b)^\ell,$$

which implies $I \equiv (I^a)^{h_\mathcal{K}} \bmod \mathcal{I}^\ell$. By definition of the class number, there exists $\gamma \in \Gamma$ such that $(I^a)^{h_\mathcal{K}} = (\gamma)$, and therefore, $j(\overline{\gamma}) \equiv I \bmod \mathcal{I}^\ell$.

The sequence is exact. It remains to prove that $\mathrm{Im}(i) = \mathrm{Ker}(j)$. We clearly have the direct inclusion. For the other inclusion, let $\overline{\gamma} \in \mathrm{Ker}(j)$, and consider $\gamma \in \Gamma$ any representative. There exists $I \in \mathcal{I}$ such that $(\gamma) = I^\ell$, which means that I^ℓ is the identity element in the class group. Since ℓ is coprime to the class number, then I is the identity element in the class group as well, and so is I^{-1}. Therefore, there exists $\gamma' \in \Gamma$ such that $I^{-1} = (\gamma')$. Consequently, $(\gamma\gamma'^\ell) = I^\ell I^{-\ell} = \mathcal{K}$ which implies that $\gamma\gamma'^\ell$ belongs to $\mathcal{O}^\times$, and thus, $\overline{\gamma}$ belongs to $\mathcal{O}^\times/(\mathcal{O}^\times)^\ell = \mathrm{Im}(i)$.

The sequence splits. For each ideal $I \in \mathcal{I}$, let $\gamma_I \in \Gamma$ a generator of $I^{h_\mathcal{K}}$. Recall Bezout's identity $ah_\mathcal{K} + b\ell = 1$ with $a, b \in \mathbb{Z}$ and define the section $s : \mathcal{I}/\mathcal{I}^\ell \to \Gamma/\Gamma^\ell$ as $s(\overline{I}) = \overline{\gamma_I^a}$. The morphism s is clearly well defined. Further, for all $\overline{I} \in \mathcal{I}/\mathcal{I}^\ell$, we have $j(s(\overline{I})) = j(\overline{\gamma_I^a}) = \overline{(\gamma_I^a)} = \overline{I^{ah_\mathcal{K}}} = \overline{I}$. Therefore $j \circ s = \mathrm{Id}$, and the sequence splits.

It remains to prove the basis statement. Direchlet's theorem states the following group isomorphism

$$\mathcal{O}^\times \simeq \mu_\mathcal{K} \times \mathbb{Z}^r.$$

Since $|\mu_\mathcal{K}|$ is coprime to ℓ, then by Bezout's identity, there exists $c, d \in \mathbb{Z}$ such that $c|\mu_\mathcal{K}| + d\ell = 1$. Then for any $\epsilon \in \mu_\mathcal{K}$, we have $\epsilon = \epsilon^{c|\mu_\mathcal{K}|+d\ell} = (\epsilon^d)^\ell$. This proves that $\mu_\mathcal{K}/\mu_\mathcal{K}^\ell$ is the trivial group. We have the following vector space isomorphism

$$\mathcal{O}^\times/(\mathcal{O}^\times)^\ell \simeq (\mathbb{Z}/\ell\mathbb{Z})^r,$$

thus proving the unique representation of units modulo $(\mathcal{O}^\times)^\ell$ in a system of fundamental units. The unique representation of ideals within $\mathcal{I}/\mathcal{I}^\ell$ in the set $\mathcal{F}/\mathcal{F}^\ell$ directly results from the unique factorization of ideals.

A.2 Proof of Proposition 3

By Dirichlet's theorem, the unit rank of $\mathcal{K}$ is $r_\mathcal{K} = m/2 - 1$ since its signature is $(0, m/2)$, and the unit rank of the subfield $\tilde{\mathcal{K}}$ is $m/2 - 1$ as well since its signature is $(m/2, 0)$. Therefore, the group of units $\mathcal{O}_\mathcal{K}^\times$ and $\mathcal{O}_{\tilde{\mathcal{K}}}^\times$ have the same rank, and hence their index v is finite. Suppose now that v is coprime to ℓ. To not encumber the notations, we write r instead of $r_\mathcal{K}$.

Let $\{u_1, \dots, u_r\}$ a system of fundamental units of $\tilde{\mathcal{K}}$. When considered modulo $(\mathcal{O}_{\tilde{\mathcal{K}}}^\times)^\ell$, this family forms an $\mathbb{F}_\ell$-basis of the vector space $\mathcal{O}_{\tilde{\mathcal{K}}}^\times/(\mathcal{O}_{\tilde{\mathcal{K}}}^\times)^\ell$. We shall prove that it is also a basis of $\mathcal{O}_\mathcal{K}^\times/(\mathcal{O}_\mathcal{K}^\times)^\ell$. Since both spaces have the same dimension, it is sufficient to prove that the family is free.

Considered as elements of $\mathcal{O}_{\mathcal{K}}^{\times}$, let $\lambda_1, \dots \lambda_r \in \mathbb{F}_\ell$ such that $u_1^{\lambda_1} \dots u_r^{\lambda_r} \equiv 1 \bmod (\mathcal{O}_{\mathcal{K}}^{\times})^\ell$. Then, there exists $\mu \in \mathcal{O}_{\mathcal{K}}^{\times}$ such that $u_1^{\lambda_1} \dots u_r^{\lambda_r} \times \mu^\ell = 1$ where the equality is in $\mathcal{O}_{\mathcal{K}}^{\times}$. Raising both sides of the equality to the power the index $v = [\mathcal{O}_{\mathcal{K}}^{\times} : \mathcal{O}_{\tilde{\mathcal{K}}}^{\times}]$, we get

$$u_1^{\lambda_1 \cdot v} \dots u_r^{\lambda_r \cdot v} \times (\mu^v)^\ell = 1.$$

Since μ^v belongs to $\mathcal{O}_{\tilde{\mathcal{K}}}^{\times}$, the above equality holds in $\mathcal{O}_{\tilde{\mathcal{K}}}^{\times}$. Therefore we have $u_1^{\lambda_1 \cdot v} \dots ur^{\lambda_r \cdot v} \equiv 1 \bmod (\mathcal{O}_{\tilde{\mathcal{K}}}^{\times})^\ell$, from which we deduce $\lambda_1 \times v \equiv \dots \equiv \lambda_r \times v \equiv 0 \bmod \ell$. Since v is coprime to ℓ, we have $\lambda_i \equiv 0 \bmod \ell$ for all $1 \leq i \leq r$, which proves that $\{u_1, \dots, u_r\} \bmod (\mathcal{O}_{\mathcal{K}})^\ell$ is a $\mathbb{F}_\ell$ basis of $\mathcal{O}_{\mathcal{K}}^{\times}/(\mathcal{O}_{\mathcal{K}}^{\times})^\ell$.

Let $u \in \mathcal{O}_{\mathcal{K}}^{\times}/(\mathcal{O}_{\mathcal{K}}^{\times})^\ell$. We just proved that u belongs to $\mathcal{O}_{\tilde{\mathcal{K}}}^{\times}/(\mathcal{O}_{\tilde{\mathcal{K}}}^{\times})^\ell$, which means that $\sigma^{k/2}(u) = u$. Applying Corollary 1 we get $\mathrm{vlog}(u) = 0$.